

pyAmpliCol 0.1.0

Methodology and Performance Report

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Abstract

This report presents the physics coverage, numerical validation, and performance of `pyAmpliCol` 0.1.0. We compare compiled, eager, and recurrence evaluation for the built-in Standard Model and for the same model imported from UFO or serialized JSON, together with scalar-contact and scalar-gravity benchmarks. Every reported timing is associated with a documented software revision, hardware profile, numerical configuration, and validation reference appropriate to the corresponding data set.

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1 Scope

`pyAmpliCol` generates and evaluates tree-level scattering amplitudes. Compiled mode specializes a shared-current directed acyclic graph (DAG) to a process. Eager mode evaluates the same graph with reusable model kernels. Recurrence mode constructs a compact current schedule directly. The three modes implement the same physics normalization, colour treatment, helicity sums, and resolved observables. The original `AmpliCol` program and its use in event generation are described in Ref. [5].

The report has three purposes:

1. define the observables and conditions required for a reproducible benchmark;
2. preserve a stable matrix of representative process families and colour approximations; and
3. present performance only after numerical agreement has been verified.

In each architecture-specific report, an N/A entry denotes a process/multiplicity combination outside the declared scope or an observable that does not apply. Every applicable cell across the complete declared multiplicity range must instead contain a validated measurement or an documented resource-limit status. A measurement that does not pass numerical validation is labelled explicitly and never replaced by a timing value. No value is copied, interpolated, or estimated from a neighbouring cell.

2 Current Recursion and Model Lowering

2.1 Shared-current DAG

For an external subset I , particle state t , and local spin or flow label χ , an off-shell current has the schematic recursion

$$J_{t,\chi}^\alpha(I) = P^\alpha_\beta(t, p_I) \sum_{I=A \dot{\cup} B} V^\beta_{\gamma\delta}(t_A, t_B \rightarrow t) J_{t_A, \chi_A}^\gamma(A) J_{t_B, \chi_B}^\delta(B), \quad p_I = \sum_{i \in I} p_i. \quad (1)$$

Current identity includes the particle, external labels, helicity ancestry, chirality or spin state, flavour and charge flow, colour state, momentum mask, and any auxiliary-current kind. Contributions with the same complete identity share one node. A topological sweep therefore evaluates each retained current once per phase-space point and reuses it in all dependent amplitudes. This is a Berends–Giele-type construction [1].

The Universal Feynman Output format (UFO; originally Universal FeynRules Output) describes particles, parameters, interactions, Lorentz tensors, colour tensors, and propagators in a model-independent form [3, 4]. `pyAmpliCol` normalizes a trusted UFO Python module, or its own serialized JSON representation of that model, to the same canonical representation used by process generation. The UFO module is executable Python; the corresponding serialized JSON form is data-only.

Optimizations of imported models are based on exact colour, Lorentz, propagator, and coupling identities rather than particle names or numerical codes. For Standard-Model validation processes, the built-in and imported routes must produce equivalent current, interaction, amplitude, and colour content. Algebraic reuse is allowed only when the relevant identity has been established exactly.

2.2 Contact terms and spin two

A supported vertex with valence above three is lowered to a deterministic tree of non-propagating auxiliary currents. The construction inserts the original coupling once and introduces no pole. Direct contraction of the original contact tensor is the validation reference for the scalar-contact ladder.

The scalar-gravity ladder exercises rank-two external states and momentum-dependent interactions. For a massless graviton, the helicity-two polarization tensor is represented as

$$\epsilon_{\pm 2}^{\mu\nu} = \epsilon_\pm^\mu \epsilon_\pm^\nu. \quad (2)$$

Validation checks symmetry, transversality, tracelessness, normalization, and agreement between ordinary and higher-precision evaluation.

3 Colour and Runtime Semantics

The process matrices use three colour contracts:

LC	Leading-colour ordered amplitudes with physical colour-flow selectors [2].
NLC	Every colour-metric entry whose leading power is at most two below the leading power of the complete basis.
Full	The complete supported sparse colour metric for the generated amplitude basis.

For NLC, “retained” applies to a colour-matrix *entry*, not to a truncated numerical coefficient. Write the exact overlap of two colour-flow tensors as

$$C_{ab}(N_c) = \sum_p c_p^{(ab)} N_c^p, \quad p_{ab} = \max\{p : c_p^{(ab)} \neq 0\}, \quad (3)$$

and let P be the leading power of the complete colour basis. The sparse NLC metric is

$$C_{ab}^{\text{NLC}} = \begin{cases} C_{ab}(3), & p_{ab} \geq P - 2, \\ 0, & p_{ab} < P - 2. \end{cases} \quad (4)$$

Thus every qualifying entry keeps all of its finite- N_c terms. For three open fundamental lines and one adjoint, $P = 4$. Representative retained off-diagonal overlaps are

exact overlap	highest power	NLC at $N_c = 3$	full at $N_c = 3$
$-N_c^3 + N_c$	3	-24	-24
$N_c^2 - 1$	2	8	8

The powers and coefficients are retained exactly. When only the upper-triangular part of the colour metric is stored, off-diagonal interference terms receive the usual factor of two; this factor is not part of the colour coefficient in Eq. (4).

The summed observable returns one matrix element per point. Leading-colour resolved values have axes (`point`, `helicity`, `flow`); NLC and full-colour values have one contracted colour component per helicity. Summing all non-point axes must reproduce the directly evaluated total. Selectors refer to physical helicities and flows.

All pyAmpliCol binary64 interfaces use the same native numerical core. Cross-interface agreement therefore checks data conventions, while a separately generated value is required for an independent physics comparison.

The JIT backend stores a portable SymJIT application for binary64 evaluation; the receiving host creates code appropriate for its processor. ASM and C++ payloads are target-specific. Evaluation above binary64 precision uses the retained symbolic evaluator state.

4 Benchmark Methodology

4.1 Configuration contract

Each measurement begins from an immutable resolved configuration. Unless a table states otherwise, measurements use a batch size of 128, two warmup runs, at least five timed samples, a five-second sampling target, and one CPU core. Compiled process matrices use JIT O3; eager and recurrence matrices use portable JIT O2 prepared models. Explicitly labelled rows in the dedicated Z ladder use JIT O1, JIT O3, ASM O3, or C++ O3. Backend-specific optimization settings are retained with the measurement record.

Generation time begins with the resolved model input required by an execution mode and ends with a validated, loadable process representation. Compiled mode includes process-specific evaluator construction and backend compilation. Eager and recurrence modes include construction and serialization of their respective numerical schedules. Preparation of a reusable built-in or UFO/JSON model bundle occurs before this timer and is therefore excluded from the process-generation values in the tables.

Runtime measurements use the summed matrix element. The reported native wall time starts after caller-language array conversion and includes source construction, numerical evaluation, selector handling, and final reduction. Caller-language packing is excluded. When a backend exposes a separately measured execution attribution, the tables show it alongside wall time; otherwise the field is marked NOT EXPOSED and is never inferred or replaced by zero. Comparisons use the same process, colour accuracy, parameters, phase-space point, precision, selectors, and normalization.

4.2 Benchmark platform

This document is an architecture-specific edition. The recorded hardware, operating system, and processor summary is pending measurement-host authentication. The measurement profile is `macbook_M3_manual`; its recorded software environment is pending an authenticated post-checkpoint build. Measurements from different processor architectures are reported separately and are never combined in a summary ratio. Unless a table states otherwise, generation and evaluation use one CPU core and a five-second sampling target.

4.3 Numerical validation

For each Standard-Model process, all compared execution modes and the original `AmpliCol` reference are evaluated at the same deterministic phase-space point with identical numerical parameters. Cross-mode comparisons use relative tolerance 10^{-12} and absolute tolerance 10^{-15} ; comparisons with the independent `AmpliCol` result use relative tolerance 10^{-8} . Scalar-model ladders instead compare binary64 evaluation with higher precision and, where applicable, direct contact-tensor contraction. Every result must reproduce its resolved sum and preserve expected structural zeros.

Timing and numerical validation are separate requirements: a timing is reported only when the corresponding matrix element passes all validation checks, and numerical agreement alone does not substitute for a measured generation or runtime value.

5 Standard-Model Process Matrices

The process matrices use recurrence as the primary `pyAmpliCol` execution mode. Built-in-SM and UFO-SM recurrence JIT O2 are compared with independent original-`AmpliCol` results. Separate built-in-SM matrices compare compiled JIT O3 and eager JIT O2 with recurrence JIT O2 for the same process, colour treatment, and observable.

The canonical matrix spans electroweak vector production, charged-current and leptonic channels, heavy-quark production, pure gluons, multi-boson final states, Higgs-associated final states, and three- and four-quark-line cases. Here n is the number of final-state particles. LC coverage retains every declared family through $n = 9$. NLC and full-colour grids retain $n = 1, \dots, 5$. A process/multiplicity combination outside its family definition is marked NOT APPLICABLE; this is table structure, not an unfilled measurement. A complete edition accounts for every applicable cell across the full declared range. Resource-limited cells carry a documented status; every successful cell is numerical and validated. An in-scope result that does not pass numerical validation is identified by a status and does not carry a timing ratio.

The UFO-SM recurrence matrices use the same process definitions, colour contracts, numerical parameters, selectors, and validation points as the built-in-SM recurrence matrices. Their source is the packaged Standard Model loaded through the serialized-JSON interface. This comparison verifies that the imported and built-in model routes implement the same requested physics. Compiled and eager process matrices are built-in-only, isolating differences between execution modes from differences in model loading.

Numerical validation summary

Only measurements whose numerical checks pass are counted below. The scope comprises every applicable report cell across the complete declared multiplicity range.

final-state multiplicity	declared	static N/A	verified	measurable coverage
1	64	0	64	verified
2	186	0	186	verified
3	206	0	206	verified
4	306	0	306	verified
5	305	0	305	verified
6	205	4	198	policy-complete (3 censored)
7	175	10	155	policy-complete (10 censored)
8	175	10	8	incomplete (2 censored, 155 open)
9	174	10	3	incomplete (4 censored, 157 open)
total	1796	34	1431	incomplete (19 censored, 312 open)

Full-catalog display accounting through $n = 9$: 1796 declared cells, comprising 1762 measurable cells and 34 catalog-authenticated static N/A rows; 432 matrix process/multiplicity positions marked NOT APPLICABLE; and 36 reference execution fields marked NOT EXPOSED. The latter two categories are intentional table structure, not unfilled measurements.

validation comparison	passed	max. relative difference
independent original-AmpliCol reference records	251	reference
pyAmpliCol versus independent reference	468	1.121×10^{-11}
compiled/eager versus recurrence	497	3.328×10^{-13}
optimized result versus resolved sum	1180	3.254×10^{-13}
scalar result versus higher precision	55	1.909×10^{-14}

Profile: macbook_M3_manual. Measured source: **source identity is incomplete or mixed**. Independent-reference comparisons use a relative tolerance of 10^{-8} ; cross-mode, resolved-sum, and higher-precision comparisons use 10^{-12} .

Workloads and denominators

For a reported timing quantity T , each multiplier is

$$r_{\text{candidate/baseline}} = \frac{T_{\text{candidate}}}{T_{\text{baseline}}}. \quad (5)$$

Green, orange, and red denote respectively $r < 1$, $1 \leq r < 2$, and $r \geq 2$. Green therefore means that the candidate is faster than the baseline named in the table heading. The recurrence matrices use original AmpliCol as the baseline. The compiled and eager matrices use built-in-SM recurrence JIT O2 as the baseline.

Every LC comparison contains two complete-coverage layouts:

- topology replay, measured with one physical colour flow selected and all retained helicities summed; and
- the all-flow union, measured with one physical helicity selected and all retained colour flows summed.

Neither selection is fixed during generation. Each LC cell reports the two baseline generation quantities and wall times, followed by candidate generation entries and runtime ratios. The muted supplementary ratio first compares separately measured execution attributions when both sides expose compatible boundaries; otherwise it compares authenticated evaluator totals. Only against original AmpliCol, which has no evaluator-total clock, may a candidate evaluator total use the legacy direct execution clock (or wall when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. If no compatible supplementary clock exists, only the wall ratio is shown. Compiled and eager generation are compared with recurrence using the same layout.

The two original-AmpliCol reference workloads have different setup boundaries. For the selected-flow workload, the reported generation quantity is creation of the process library. For the all-flow workload, it is preparation of direct all-flow evaluation. Runtime ratios compare the same physical observable. The all-flow setup quantities are not like-for-like, so the table shows the candidate process-generation time as an absolute value and forms no ratio.

NLC and full colour use one helicity-summed, colour-contracted calculation per mode. Their cells report the absolute baseline generation and wall time, followed by the candidate generation and runtime ratios. No per-flow scalar is defined for these contracted colour accuracies.

The wall quantity is the repeated native evaluation time after caller-language input conversion and is the primary runtime observable. Independent accumulated evaluator-total and narrower recurrence-core clocks remain in the detailed measurement evidence; neither is copied from wall time or from the other. The compact runtime notation (**[xS]xW**) shows a compatible supplementary clock multiplier **xS** in muted brackets and the primary wall multiplier **xW** in its performance colour. The hierarchy above determines whether **xS** is execution attribution or a diagnostic evaluator-total fallback. If no supplementary ratio is eligible, the cell shows only (**xW**).

Preparation of reusable model-level kernels is excluded from process-generation time. Compiled generation still includes process-specific evaluator construction and backend compilation.

A timing is shown only after recurrence agrees with original AmpliCol at the independent-reference tolerance and compiled and eager results agree with recurrence at the stricter cross-mode tolerance. Built-in and UFO-SM recurrence results are also compared directly at that stricter tolerance, rather than relying on their separate agreement with AmpliCol. Each result must reproduce its resolved sum. For LC, the common flow-helicity component is compared directly between topology replay and all-flow union, and every pyAmpliCol all-flow result is also compared with the matching independently generated AmpliCol component. A failed validation has no timing ratio.

Cell and summary layout

contract	baseline entry	candidate entries
LC, upper	topology replay all-flow generation quantities in seconds	layout-matched ratios, except that the all-flow candidate against original AmpliCol is shown absolutely without a ratio
LC, lower NLC and full-colour, upper	selected-flow all-flow wall times in $\mu s/pt$ contracted generation time in seconds	compact ([xS]xW) comparisons, or (xW) when only wall is comparable candidate generation ratio
NLC and full-colour, lower	contracted wall time in $\mu s/pt$	compact ([xS]xW) comparison, or (xW) when only wall is comparable

Summary rows contain multipliers only, ordered as minimum, maximum, median, arithmetic mean, and weighted mean. The arithmetic mean averages the individual candidate/baseline multipliers; the weighted mean is the ratio of sums $\sum_i T_{\text{candidate},i} / \sum_i T_{\text{baseline},i}$. Generation summaries use topology replay for LC and the contracted workload for NLC and full colour; the non-comparable LC all-flow generation quantity is omitted. Runtime summaries use wall time only, with separate aligned LC topology-replay and all-flow lines. Structurally inapplicable, resource-limited, unsupported, and validation-failed cells do not contribute.

The first three matrices provide a built-in-SM overview against original **AmpliCol**. For each process, multiplicity, colour treatment, and LC workload, they select the validated pyAmpliCol mode with the smallest wall time. Selection is independent in every cell: the overview records the measured winner rather than prescribing one mode for all processes. Its upper row reports generation and its lower row reports runtime. LC pairs use non-union-flow | union-flow ordering. Since original-**AmpliCol** all-flow setup and py**AmpliCol** process generation have different boundaries, the latter is shown absolutely rather than as a misleading ratio.

5.1 Built-in SM best measured mode versus AmpliCol: LC

Built-in SM best measured mode versus AmpliCol LC (block 1 of 3)

ID	base process	metric	n=1					n=2					n=3								
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.14		5.50e-05	(r) x2.11		1.97 (c)	1.96		8.50e-05	(c) x1.25		2.19 (c)	2.19		0.000263	(c) x1.63		2.76 (c)	
		run [μ s/pt]	0.0218		0.0473	([x5.74] x7.73)		([x1.83] x1.83)	0.127		0.111	([x3.49] x3.49)		([x1.41] x1.41)	0.461		0.328	([x1.64] x1.64)		([x0.945] x0.945)	
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	1.93		5.60e-05	(r) x2.25		4.35 (r)	1.95		8.10e-05	(r) x2.27		2.05 (c)	2.07		0.000260	(c) x1.42		2.29 (c)	
		run [μ s/pt]	0.0133		0.0467	([x5.91] x8.80)		([x0.796] x2.00)	0.0734		0.115	([x3.13] x4.09)		([x1.37] x1.37)	0.267		0.317	([x1.90] x1.90)		([x0.981] x0.981)	
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE					2.16		7.60e-05	2.78 (c)		2.54 (c)	2.26		0.000222	3.18 (c)		2.80 (c)		
		run [μ s/pt]						0.0691		0.118	0.240		0.161	0.177		0.196	0.432		0.229		
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE					2.06		7.40e-05	(c) x1.08		2.07 (c)	2.09		0.000218	(r) x2.15		2.04 (c)		
		run [μ s/pt]						0.0331		0.0899	([x3.90] x3.90)		([x1.58] x1.58)	0.0638		0.160	([x2.26] x3.63)		([x1.30] x1.30)		
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE					2.05		7.50e-05	(c) x1.19		2.47 (c)	2.16		0.000227	(c) x1.36		2.34 (c)		
		run [μ s/pt]						0.159		0.115	([x3.05] x3.05)		([x1.38] x1.38)	0.718		0.243	([x1.66] x1.66)		([x1.12] x1.12)		
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE					2.17		8.70e-05	(c) x1.14		2.13 (c)	3.24		0.000340	(c) x1.62		2.86 (c)		
		run [μ s/pt]						0.102		0.155	([x3.85] x3.85)		([x1.30] x1.30)	0.704		0.639	([x1.63] x1.63)		([x1.03] x1.03)		
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE					2.02		8.30e-05	(c) x1.29		2.25 (c)	2.51		0.000257	(c) x1.63		3.17 (c)		
		run [μ s/pt]						0.0501		0.112	([x5.19] x5.19)		([x1.43] x1.43)	0.242		0.379	([x2.44] x2.44)		([x0.964] x0.964)		
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	NOT APPLICABLE					2.85		0.000112	(c) x1.05		2.50 (c)	2.66		0.000563	(c) x2.58		3.41 (c)		
		run [μ s/pt]						0.187		0.250	([x1.77] x1.77)		([x0.962] x0.962)	0.970		1.54	([x1.38] x1.38)		([x0.778] x0.779)		
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					3.60		0.000246	(c) x1.03		2.40 (c)			
		run [μ s/pt]											1.54		0.390	([x1.75] x1.75)		([x0.910] x0.910)			
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE								
		run [μ s/pt]																			
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE								
		run [μ s/pt]																			
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE								
		run [μ s/pt]																			
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE								
		run [μ s/pt]																			
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE								
		run [μ s/pt]																			
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE								
		run [μ s/pt]																			
			n=1 [r:2 c:0 e:0]					n=2 [r:1 c:6 e:0]					n=3 [r:1 c:7 e:0]								
summary: generation			x2.11		x2.25		x2.18		x2.18		x2.17		x1.05		x2.27		x1.19		x1.33		x1.30
summary: run single-flow, hel. sum			x7.73		x8.80		x8.26		x8.26		x8.13		x1.77		x5.19		x3.85		x3.62		x3.20
summary: run all-flows, single hel.			x1.83		x2.00		x1.92		x1.92		x1.92		x0.962		x1.58		x1.38		x1.35		x1.28
													x0.779		x1.30		x0.973		x1.00		x0.920

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μ s/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. For the LC all-flow workload, AmpliCol direct-evaluation setup and pyAmpliCol process generation have different boundaries, so the candidate generation value is shown absolutely; no misleading multiplier is formed. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

Built-in SM best measured mode versus AmpliCol LC (block 2 of 3)

ID	base process	metric		n=4					n=5					n=6																	
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.08		0.00124	(c)	x3.02		4.78	(c)	2.34		0.00975	(r)	x2.35		15.9	(c)	2.95		0.0884	(r)	x2.63		99.5	(c)					
		run [μ s/pt]	1.46		1.19	([x1.42]	x1.42)		([x0.813]	x0.813)	4.56		6.53	([x1.32]	x1.40)		([x0.733]	x0.733)	13.6		41.4	([x1.11]	x1.15)		([x0.878]	x0.878)					
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	2.15		0.00120	(c)	x2.27		3.47	(c)	2.15		0.00981	(c)	x6.38		9.76	(c)	2.41		0.0872	(r)	x2.78		62.6	(c)					
		run [μ s/pt]	0.922		1.19	([x1.54]	x1.54)		([x0.737]	x0.737)	2.92		6.57	([x1.53]	x1.53)		([x0.734]	x0.735)	9.32		42.1	([x1.14]	x1.18)		([x0.907]	x0.907)					
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	2.16		0.000960	(c)	x2.46		3.87	(c)	2.40		0.00710	(c)	x3.85		6.46	(c)	2.81		0.0621	(c)	x9.89		22.3	(c)					
		run [μ s/pt]	0.450		0.416	([x1.66]	x1.66)		([x0.976]	x0.977)	1.37		1.38	([x1.37]	x1.37)		([x0.718]	x0.718)	4.08		7.30	([x1.40]	x1.40)		([x0.732]	x0.733)					
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	2.20		0.00102	(c)	x1.39		2.37	(c)	2.22		0.00727	(c)	x2.19		3.59	(c)	2.46		0.0800	(c)	x5.13		9.73	(c)					
		run [μ s/pt]	0.162		0.360	([x2.51]	x2.51)		([x1.02]	x1.02)	0.626		1.23	([x1.69]	x1.69)		([x0.805]	x0.805)	2.23		7.06	([x1.57]	x1.57)		([x0.712]	x0.712)					
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	2.41		0.00107	(c)	x2.03		3.10	(c)	3.53		0.00791	(c)	x3.17		6.79	(c)	5.24		0.0679	(c)	x8.38		29.9	(c)					
		run [μ s/pt]	2.12		0.686	([x1.40]	x1.40)		([x0.788]	x0.788)	6.60		2.56	([x1.20]	x1.20)		([x0.630]	x0.631)	18.1		13.5	([x1.29]	x1.29)		([x0.617]	x0.617)					
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.22		0.00198	(c)	x2.57		7.30	(c)	23.1		0.0186	(c)	x4.17		7.66	(r)	96.5		0.208	(r)	x1.29		28.7	(r)					
		run [μ s/pt]	2.46		3.34	([x1.60]	x1.60)		([x1.19]	x1.19)	8.19		25.6	([x1.55]	x1.55)		([x1.04]	x1.07)	23.0		198	([x1.41]	x1.45)		([x1.14]	x1.15)					
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	3.43		0.00123	(c)	x3.26		5.57	(c)	6.72		0.00991	(c)	x5.06		19.8	(c)	11.2		0.0958	(c)	x3.40		117	(c)					
		run [μ s/pt]	0.895		1.56	([x1.56]	x1.56)		([x0.789]	x0.789)	3.03		9.64	([x1.54]	x1.54)		([x0.697]	x0.697)	9.31		65.5	([x1.52]	x1.52)		([x0.787]	x0.787)					
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]	3.20		0.00393	(r)	x2.96		13.1	(c)	4.60		0.0636	(r)	x5.37		20.4	(r)	10.7		2.54	(r)	x7.12		192	(r)					
		run [μ s/pt]	3.32		12.4	([x1.16]	x1.24)		([x0.746]	x0.746)	11.1		106	([x0.965]	x0.997)		([x0.881]	x0.888)	39.3		1.18e+03	([x0.747]	x0.761)		([x1.37]	x1.36)					
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.13		0.00127	(c)	x1.08		2.86	(c)	9.65		0.00703	(c)	x1.75		8.23	(c)	23.4		0.0670	(c)	x1.44		14.8	(c)					
		run [μ s/pt]	4.88		0.680	([x1.44]	x1.44)		([x0.844]	x0.844)	13.9		1.69	([x2.34]	x2.34)		([x0.758]	x0.758)	37.3		6.49	([x1.15]	x1.15)		([x0.560]	x0.560)					
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	10.3		0.000925	(c)	x0.394		3.27	(c)	13.3		0.00690	(c)	x0.377		3.82	(c)	15.7		0.0574	(c)	x0.606		5.59	(c)					
		run [μ s/pt]	0.769		0.579	([x1.71]	x1.71)		([x0.828]	x0.828)	1.64		0.843	([x1.62]	x1.62)		([x0.881]	x0.881)	3.92		1.67	([x1.38]	x1.38)		([x0.731]	x0.732)					
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	6.62		0.000955	(c)	x0.707		3.00	(c)	15.6		0.00714	(c)	x0.807		4.89	(c)	36.8		0.0632	(c)	x1.38		13.9	(c)					
		run [μ s/pt]	1.11		0.717	([x1.82]	x1.82)		([x0.959]	x0.960)	3.29		1.91	([x1.81]	x1.81)		([x0.960]	x0.960)	10.5		8.25	([x1.67]	x1.67)		([x0.821]	x0.821)					
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	11.2		0.000994	(r)	x0.435		10.6	(c)	17.2		0.00704	(c)	x0.580		8.50	(c)	17.8		0.0571	(c)	x1.03		11.7	(c)					
		run [μ s/pt]	1.50		0.866	([x1.71]	x1.90)		([x0.790]	x0.790)	3.06		1.15	([x1.03]	x1.03)		([x0.749]	x0.750)	6.69		2.14	([x0.823]	x0.824)		([x0.646]	x0.646)					
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	2.93		0.00104	(c)	x2.46		5.70	(c)	6.20		0.00749	(c)	x3.56		14.5	(c)	13.0		0.0626	(c)	x9.59		58.9	(c)					
		run [μ s/pt]	0.171		0.874	([x3.30]	x3.30)		([x0.850]	x0.850)	0.435		3.68	([x2.41]	x2.41)		([x0.794]	x0.794)	1.15		24.2	([x2.71]	x2.71)		([x0.582]	x0.582)					
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					STATIC N/A					STATIC N/A					9.13 (r)			45.9 (c)					
		run [μ s/pt]	NOT APPLICABLE					NOT APPLICABLE					STATIC N/A					STATIC N/A					3.64			7.83					
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	2.59		0.00100	(c)	x5.22		8.54	(c)	4.89		0.00784	(c)	x8.94		23.6	(c)	8.98		0.0659	(c)	x18.4		97.5	(c)					
		run [μ s/pt]	0.268		1.33	([x2.62]	x2.62)		([x0.922]	x0.923)	0.600		6.17	([x1.95]	x1.96)		([x0.806]	x0.806)	1.40		36.2	([x2.32]	x2.32)		([x0.662]	x0.662)					
summary: generation			n=4 [r:2 c:12 e:0]					n=5 [r:2 c:12 e:0]					n=6 [r:4 c:10 e:0]																		
summary: run single-flow, hel. sum			x0.394		x5.22		x2.36		x2.16		x1.63		x0.377		x8.94		x3.36		x3.47		x2.72		x0.606		x18.4		x3.09		x5.22		x2.96
summary: run all-flows, single hel.			x1.24		x3.30		x1.63		x1.84		x1.54		x0.997		x2.41		x1.54		x1.60		x1.57		x0.761		x2.71		x1.39		x1.46		x1.19
			x0.737		x1.19		x0.836		x0.876		x0.841		x0.631		x1.07		x0.776		x0.802		x0.880		x0.560		x1.36		x0.732		x0.796		x1.24

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μ s/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. For the LC all-flow workload, AmpliCol direct-evaluation setup and pyAmpliCol process generation have different boundaries, so the candidate generation value is shown absolutely; no misleading multiplier is formed. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

Built-in SM best measured mode versus AmpliCol LC (block 3 of 3)

ID	base process	metric		n=7				n=8				n=9			
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	4.68	1.08	(r) x3.23	1.14e+03 (c)	8.94	13.3	N/A	N/A	26.1	>15GB	N/A	N/A	
		run [μ s/pt]	38.9	323	([x0.991] x1.01)	([x1.38] x1.38)	95.8	6.18e+03	N/A	N/A	228	>15GB	N/A	N/A	
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	3.08	1.00	(r) x3.80	58.5 (r)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	27.5	310	([x0.972] x0.997)	([x1.55] x1.56)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	3.35	0.631	(r) x2.16	133 (c)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	12.1	45.3	([x1.11] x1.16)	([x0.851] x0.851)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	2.72	0.643	(r) x2.34	58.5 (c)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	8.99	43.4	([x0.961] x0.997)	([x0.855] x0.855)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	11.3	0.715	(r) x1.39	237 (c)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	47.3	84.7	([x1.16] x1.18)	([x0.676] x0.676)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	645	4.39	(r) x0.717	373 (r)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	61.4	3.11e+03	([x1.42] x1.45)	([x1.44] x1.46)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	36.0	1.08	(r) x1.87	1.05e+03 (c)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	27.0	509	([x1.41] x1.47)	([x1.10] x1.10)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]	30.5	209	(r) x13.7	2.88e+03 (r)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	86.1	2.85e+04	([x0.847] x0.884)	([x1.47] x1.49)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	58.0	0.622	(r) x0.378	85.0 (c)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	83.8	28.9	([x1.28] x1.28)	([x0.516] x0.517)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	22.4	0.572	(c) x0.915	13.2 (c)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	9.89	5.34	([x1.24] x1.24)	([x0.634] x0.634)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	120	0.617	(c) x1.55	66.4 (c)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	29.7	44.9	([x1.58] x1.58)	([x0.804] x0.804)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	25.0	0.610	(c) x1.54	27.7 (c)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	16.1	6.26	([x0.765] x0.765)	([x0.593] x0.593)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	29.5	0.656	(c) x6.52	317 (c)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	2.95	142	([x2.51] x2.51)	([x0.671] x0.671)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	STATIC N/A	STATIC N/A	55.9 (r)	210 (c)	STATIC N/A	STATIC N/A	N/A	N/A	STATIC N/A	STATIC N/A	N/A	N/A	
		run [μ s/pt]	STATIC N/A	STATIC N/A	7.17	38.1	STATIC N/A	STATIC N/A	N/A	N/A	STATIC N/A	STATIC N/A	N/A	N/A	
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	18.5	0.684	(c) x12.6	600 (c)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	3.56	402	([x2.75] x2.75)	([x0.510] x0.510)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
summary: generation		x0.378		x13.7	x2.01	x3.77	x1.68	n=8		N/A		n=9		N/A	
summary: run single-flow, hel. sum		x0.765		x2.75	x1.21	x1.38	x1.20	N/A		N/A		N/A		N/A	
summary: run all-flows, single hel.		x0.510		x1.56	x0.827	x0.937	x1.46	N/A		N/A		N/A		N/A	

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μ s/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. For the LC all-flow workload, AmpliCol direct-evaluation setup and pyAmpliCol process generation have different boundaries, so the candidate generation value is shown absolutely; no misleading multiplier is formed. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

5.2 Built-in SM best measured mode versus AmpliCol: NLC

Built-in SM best measured mode versus AmpliCol NLC (block 1 of 2)

ID	base process	metric	n=1					n=2					n=3				
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	1.94	(c)	x1.08			1.23	(c)	x1.97			1.96	(c)	x1.51		
		run [μ s/pt]	0.0837	([x1.73]	x1.73)			0.245	([x1.33]	x1.33)			1.35	([x0.894]	x0.895)		
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	1.93	(r)	x2.42			1.97	(c)	x1.13			2.02	(c)	x1.21		
		run [μ s/pt]	0.0520	([x1.62]	x2.19)			0.134	([x2.00]	x2.00)			0.763	([x1.08]	x1.08)		
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]			NOT APPLICABLE			2.02	(c)	x1.23			2.13	(c)	x1.33		
		run [μ s/pt]						0.178	([x1.14]	x1.14)			0.404	([x0.922]	x0.922)		
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]			NOT APPLICABLE			2.09	(r)	x2.13			2.10	(r)	x2.17		
		run [μ s/pt]						0.0974	([x0.815]	x1.20)			0.189	([x0.715]	x1.05)		
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]			NOT APPLICABLE			2.05	(c)	x1.19			2.18	(c)	x1.18		
		run [μ s/pt]						0.334	([x1.18]	x1.18)			0.992	([x1.04]	x1.04)		
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]			NOT APPLICABLE			2.20	(c)	x1.03			3.23	(c)	x1.17		
		run [μ s/pt]						0.680	([x0.881]	x0.882)			6.79	([x0.831]	x0.832)		
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]			NOT APPLICABLE			2.07	(e)	x1.03			3.51	(r)	x1.38		
		run [μ s/pt]						0.492	([x0.862]	x0.862)			2.70	([x0.697]	x0.830)		
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]			NOT APPLICABLE			2.78	(c)	x0.879			2.87	(c)	x1.26		
		run [μ s/pt]						3.79	([x0.402]	x0.402)			48.0	([x0.629]	x0.629)		
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE			3.59	(c)	x0.902		
		run [μ s/pt]											1.95	([x1.26]	x1.26)		
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE		
		run [μ s/pt]															
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE		
		run [μ s/pt]															
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE		
		run [μ s/pt]															
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE		
		run [μ s/pt]															
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE		
		run [μ s/pt]															
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE		
		run [μ s/pt]															
summary: generation			n=1 [r:1 c:1 e:0]					n=2 [r:1 c:6 e:1]					n=3 [r:2 c:7 e:0]				
summary: wall			x1.08	x2.42	x1.75	x1.75	x1.75	x0.879	x2.13	x1.16	x1.32	x1.27	x0.902	x2.17	x1.26	x1.35	x1.31
			x1.73	x2.19	x1.96	x1.96	x1.91	x0.402	x2.00	x1.16	x1.12	x0.648	x0.629	x1.26	x0.922	x0.948	x0.700

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μ s/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

Built-in SM best measured mode versus AmpliCol NLC (block 2 of 2)

ID	base process	metric	n=4			n=5			n=6		
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.09	(c)	x2.85	2.27	(r)	x12.0	NOT APPLICABLE		
		run [μ s/pt]	11.9	([x0.932] x0.932)		145	([x0.774] x0.867)				
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	2.06	(r)	x2.74	2.12	(r)	x8.44	NOT APPLICABLE		
		run [μ s/pt]	7.03	([x0.853] x0.975)		90.7	([x0.660] x0.744)				
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	2.35	(c)	x1.51	2.57	(c)	x2.63	NOT APPLICABLE		
		run [μ s/pt]	2.08	([x0.546] x0.546)		16.7	([x0.578] x0.578)				
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	2.15	(c)	x1.24	2.53	(c)	x1.56	NOT APPLICABLE		
		run [μ s/pt]	0.964	([x0.658] x0.658)		8.52	([x0.577] x0.577)				
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	2.93	(c)	x1.46	3.21	(c)	x4.97	NOT APPLICABLE		
		run [μ s/pt]	5.65	([x1.00] x1.00)		48.8	([x1.07] x1.07)				
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.25	(c)	x3.13	23.0	(e)	x6.51	NOT APPLICABLE		
		run [μ s/pt]	92.0	([x0.802] x0.802)		1.49e+03	([x0.955] x0.955)				
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	5.04	(r)	x1.41	9.64	(r)	x3.23	NOT APPLICABLE		
		run [μ s/pt]	24.8	([x0.596] x0.666)		309	([x0.711] x0.815)				
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]	4.08	(r)	x2.63	9.11	(r)	x3.14	NOT APPLICABLE		
		run [μ s/pt]	801	([x0.516] x0.645)		1.53e+04	([x0.547] x0.657)				
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.26	(c)	x0.849	21.4	(r)	x0.504	NOT APPLICABLE		
		run [μ s/pt]	5.69	([x1.17] x1.17)		44.3	([x1.14] x1.17)				
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	13.1	(c)	x0.282	14.2	(c)	x0.318	NOT APPLICABLE		
		run [μ s/pt]	1.45	([x0.860] x0.860)		3.19	([x0.797] x0.797)				
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	8.12	(r)	x0.654	20.7	(r)	x0.344	NOT APPLICABLE		
		run [μ s/pt]	4.53	([x1.18] x1.26)		26.5	([x1.06] x1.10)				
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	12.3	6.25 (c)		17.5	8.59 (c)		NOT APPLICABLE		
		run [μ s/pt]	2.90	1.37		6.22	2.74				
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	0.00110	5.14 (r)		0.0115	7.32 (r)		0.102	(r) x407	
		run [μ s/pt]	57.2	2.07		501	15.5		6.18e+03	([x0.0290] x0.0326)	
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE			NOT APPLICABLE			STATIC N/A	17.6 (r)	
		run [μ s/pt]							STATIC N/A	43.7	
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	0.00106	5.07 (r)		0.00879	(r) x1.17e+03		0.103	(c) x2.10e+03	
		run [μ s/pt]	84.1	3.30		820	([x0.0280] x0.0334)		9.90e+03	([x0.0476] x0.0476)	
summary: generation summary: wall			n=4 [r:4 c:7 e:0]			n=5 [r:7 c:4 e:1]			n=6 [r:1 c:1 e:0]		
			x0.282 x3.13 x1.46 x1.71 x1.40	x0.318 x1.17e+03 x3.19 x101 x2.83	x407 x2.10e+03 x1.25e+03 x1.25e+03 x1.26e+03						
			x0.546 x1.26 x0.860 x0.865 x0.675	x0.0334 x1.17 x0.806 x0.780 x0.661	x0.0326 x0.0476 x0.0401 x0.0401 x0.0418						

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μ s/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

5.3 Built-in SM best measured mode versus AmpliCol: full-colour

Built-in SM best measured mode versus AmpliCol full-colour (block 1 of 2)

ID	base process	metric	n=1					n=2					n=3				
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.25	(c)	x1.04			1.94	(c)	x1.27			2.21	(c)	x1.27		
		run [μ s/pt]	0.0993	([x1.45]	x1.45)			0.275	([x1.20]	x1.20)			1.60	([x0.790]	x0.790)		
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	2.00	(r)	x2.16			2.04	(r)	x2.14			1.96	(c)	x1.22		
		run [μ s/pt]	0.0595	([x1.38]	x1.83)			0.152	([x1.49]	x1.86)			0.901	([x0.899]	x0.899)		
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]			NOT APPLICABLE			2.18	2.66	(c)			2.28	3.05	(c)		
		run [μ s/pt]						0.211	0.213				0.457	0.393			
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]			NOT APPLICABLE			2.10	(r)	x2.14			2.14	(r)	x2.10		
		run [μ s/pt]						0.107	([x0.741]	x1.09)			0.209	([x0.703]	x0.990)		
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]			NOT APPLICABLE			2.10	(c)	x1.07			2.65	(c)	x1.14		
		run [μ s/pt]						0.375	([x1.04]	x1.04)			1.14	([x0.918]	x0.918)		
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]			NOT APPLICABLE			2.12	(c)	x1.10			3.93	(c)	x0.990		
		run [μ s/pt]						0.831	([x0.735]	x0.735)			8.55	([x0.733]	x0.733)		
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]			NOT APPLICABLE			2.06	(e)	x1.02			3.57	(r)	x1.33		
		run [μ s/pt]						0.622	([x0.680]	x0.680)			3.19	([x0.608]	x0.717)		
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]			NOT APPLICABLE			4.63	(c)	x0.544			2.88	(c)	x1.36		
		run [μ s/pt]						5.93	([x0.307]	x0.307)			61.4	([x0.498]	x0.498)		
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE			4.10	(c)	x0.781		
		run [μ s/pt]											2.09	([x1.18]	x1.18)		
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE		
		run [μ s/pt]															
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE		
		run [μ s/pt]															
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE		
		run [μ s/pt]															
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE		
		run [μ s/pt]															
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE		
		run [μ s/pt]															
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]			NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE		
		run [μ s/pt]															
summary: generation			n=1 [r:1 c:1 e:0]					n=2 [r:2 c:4 e:1]					n=3 [r:2 c:6 e:0]				
summary: wall			x1.04	x2.16	x1.60	x1.60	x1.57	x0.544	x2.14	x1.10	x1.33	x1.21	x0.781	x2.10	x1.25	x1.27	x1.22
			x1.45	x1.83	x1.64	x1.64	x1.60	x0.307	x1.86	x1.04	x0.987	x0.479	x0.498	x1.18	x0.845	x0.840	x0.568

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μ s/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

Built-in SM best measured mode versus AmpliCol full-colour (block 2 of 2)

ID	base process	metric		n=4		n=5		n=6					
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.07	(c)	x2.91	2.49	(r)	x11.3	NOT APPLICABLE				
		run [μ s/pt]	14.2	([x0.786] x0.786)		203	([x0.551] x0.656)						
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	2.15	(c)	x2.19	2.16	(r)	x8.19	NOT APPLICABLE				
		run [μ s/pt]	8.30	([x0.860] x0.860)		116	([x0.514] x0.586)						
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	2.46	3.73 (c)		2.55	(c)	x2.66	NOT APPLICABLE				
		run [μ s/pt]	2.40	1.16		20.1	([x0.476] x0.476)						
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	2.18	(c)	x1.18	2.36	(c)	x1.68	NOT APPLICABLE				
		run [μ s/pt]	1.12	([x0.557] x0.557)		10.2	([x0.479] x0.479)						
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	2.51	(c)	x1.66	3.24	(c)	x4.80	NOT APPLICABLE				
		run [μ s/pt]	6.06	([x0.921] x0.921)		55.4	([x0.924] x0.924)						
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.27	(c)	x3.11	21.6	(e)	x7.10	NOT APPLICABLE				
		run [μ s/pt]	126	([x0.638] x0.638)		2.82e+03	([x0.673] x0.673)						
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	4.56	(r)	x1.59	11.2	(r)	x2.93	NOT APPLICABLE				
		run [μ s/pt]	32.9	([x0.448] x0.501)		555	([x0.401] x0.474)						
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]	4.11	(r)	x2.62	7.79	(r)	x4.22	NOT APPLICABLE				
		run [μ s/pt]	1.38e+03	([x0.303] x0.397)		5.03e+04	([x0.156] x0.240)						
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.02	(c)	x0.894	9.64	(r)	x1.09	NOT APPLICABLE				
		run [μ s/pt]	5.97	([x1.11] x1.11)		32.7	([x1.04] x1.12)						
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	11.4	(c)	x0.375	14.3	(c)	x0.310	NOT APPLICABLE				
		run [μ s/pt]	1.50	([x0.831] x0.831)		3.30	([x0.772] x0.772)						
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	7.81	(r)	x0.625	20.3	(r)	x0.345	NOT APPLICABLE				
		run [μ s/pt]	5.00	([x1.07] x1.13)		28.6	([x0.970] x1.01)						
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	12.0	6.35 (c)		16.1	8.07 (c)		NOT APPLICABLE				
		run [μ s/pt]	2.93	1.37		6.29	2.66						
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	0.00106	4.91 (r)		0.00894	7.83 (r)		0.120 (c) x1.12e+03				
		run [μ s/pt]	59.5	2.06		536	16.0		7.98e+03 ([x0.0386] x0.0386)				
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE			NOT APPLICABLE			STATIC N/A 17.7 (r)				
		run [μ s/pt]							STATIC N/A 45.3				
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	0.00147	4.94 (r)		0.00866	10.2 (r)		0.106 (c) x2.13e+03				
		run [μ s/pt]	86.2	3.22		842	27.0		1.06e+04 ([x0.0463] x0.0463)				
summary: generation				n=4 [r:3 c:7 e:0]				n=5 [r:6 c:4 e:1]				n=6 [r:0 c:2 e:0]	
				x0.375 x3.11 x1.62 x1.71 x1.46				x0.310 x11.3 x2.93 x4.06 x3.21				x1.12e+03 x2.13e+03 x1.62e+03 x1.62e+03 x1.59e+03	
summary: wall				x0.397 x1.13 x0.809 x0.773 x0.432				x0.240 x1.12 x0.656 x0.673 x0.269				x0.0386 x0.0463 x0.0425 x0.0425 x0.0430	

The candidate is selected independently in each cell and workload by the smallest validated wall time. The upper row is generation in seconds and the lower row is runtime in μ s/pt. In LC cells, each pair is non-union-flow | union-flow. Generation multipliers are coloured directly. Each runtime workload uses ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it uses authenticated evaluator totals. Because original AmpliCol exposes no evaluator-total clock, a candidate total then uses the AmpliCol direct execution clock, or its wall clock when direct execution is absent, as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; execution/execution remains the compatible supplementary attribution. The coloured xW is always the candidate/AmpliCol wall-time multiplier used for selection and as the primary figure of merit. If no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original-AmpliCol rows beyond its three-open-quark-line scope are catalog static N/A entries; their candidates are shown absolutely and excluded from ratio summaries. Unavailable baselines keep their status; candidates remain absolute. Unverified compiled or eager diagnostics are never eligible for best-mode selection or summaries and remain visibly marked unverified. Only exact stored same-point links enter ratios. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. Muted (r), (c), and (e) labels on the generation row identify the recurrence, compiled, and eager winner selected by wall time for each workload; the runtime row does not repeat them. LC generation uses non-union flow only, while LC runtime keeps separate non-union and union wall-only lines. Summary mode counts use r|c|e for recurrence JIT O2, compiled JIT O3, and eager-DAG JIT O2, respectively.

The detailed tables are grouped by execution mode, model source, and colour treatment.

5.4 Built-in SM recurrence versus AmpliCol LC

Built-in SM recurrence versus AmpliCol LC (block 1 of 3)

ID	base process	metric	n=1					n=2					n=3																
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.14		5.50e-05	x2.11		4.24	1.96		8.50e-05	x2.28		4.25	2.19		0.000263	x2.25		5.08									
		run [μ s/pt]	0.0218		0.0473	([x5.74] x7.73)		([x0.777] x1.96)	0.127		0.111	([x3.00] x3.71)		([x1.47] x2.33)	0.461		0.328	([x2.51] x2.82)		([x2.12] x2.57)									
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	1.93		5.60e-05	x2.25		4.35	1.95		8.10e-05	x2.27		4.40	2.07		0.000260	x2.11		4.47									
		run [μ s/pt]	0.0133		0.0467	([x5.91] x8.80)		([x0.796] x2.00)	0.0734		0.115	([x3.13] x4.09)		([x1.46] x2.25)	0.267		0.317	([x2.66] x3.12)		([x2.20] x2.67)									
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE					2.16		7.60e-05	4.49		4.45	2.26		0.000222	4.45		4.41										
		run [μ s/pt]						0.0691		0.118	0.295		0.243	0.177		0.196	0.512		0.422										
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE					2.06		7.40e-05	x2.14		4.45	2.09		0.000218	x2.15		4.48										
		run [μ s/pt]						0.0331		0.0899	([x2.42] x4.09)		([x1.17] x2.10)	0.0638		0.160	([x2.26] x3.63)		([x1.41] x2.14)										
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE					2.05		7.50e-05	x2.05		4.04	2.16		0.000227	x2.03		4.23										
		run [μ s/pt]						0.159		0.115	([x3.26] x3.73)		([x1.67] x2.45)	0.718		0.243	([x2.69] x2.92)		([x2.14] x2.71)										
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE					2.17		8.70e-05	x2.02		4.15	3.24		0.000340	x1.66		4.43										
		run [μ s/pt]						0.102		0.155	([x9.22] x10.2)		([x1.86] x2.58)	0.704		0.639	([x3.13] x3.37)		([x2.97] x3.34)										
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE					2.02		8.30e-05	x2.12		4.21	2.51		0.000257	x2.06		4.83										
		run [μ s/pt]						0.0501		0.112	([x4.55] x5.78)		([x1.17] x1.98)	0.242		0.379	([x4.70] x5.29)		([x2.49] x3.05)										
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	NOT APPLICABLE					2.85		0.000112	x1.68		4.55	2.66		0.000563	x2.08		4.50										
		run [μ s/pt]						0.187		0.250	([x7.64] x8.12)		([x5.35] x6.09)	0.970		1.54	([x2.94] x3.08)		([x2.09] x2.37)										
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					3.60		0.000246	x1.37		4.26											
		run [μ s/pt]											1.54		0.390	([x2.25] x2.41)		([x1.84] x2.23)											
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE																
		run [μ s/pt]																											
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE																
		run [μ s/pt]																											
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE																
		run [μ s/pt]																											
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE																
		run [μ s/pt]																											
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE																
		run [μ s/pt]																											
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE																
		run [μ s/pt]																											
summary: generation			x2.11		x2.25		x2.18		x2.18		x2.17	x1.68		x2.28		x2.12		x2.08		x2.06	x1.37		x2.25		x2.07		x1.96		x1.91
summary: run single-flow, hel. sum			x7.73		x8.80		x8.26		x8.26		x8.13	x3.71		x10.2		x4.09		x5.68		x5.95	x2.41		x5.29		x3.10		x3.33		x2.98
summary: run all-flows, single hel.			x1.96		x2.00		x1.98		x1.98		x1.98	x1.98		x6.09		x2.33		x2.83		x3.31	x2.14		x3.34		x2.62		x2.64		x2.63

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time. Its setup boundary differs from the reference, so no generation ratio is formed. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM recurrence versus AmpliCol LC (block 2 of 3)

ID	base process	metric		n=4					n=5					n=6						
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.08		0.00124	x2.31		4.33	2.34		0.00975	x2.35		4.95	2.95		0.0884	x2.63		9.39
		run [μ s/pt]	1.46		1.19	([x2.00] x2.13)		([x3.20] x3.44)	4.56		6.53	([x1.32] x1.40)		([x1.19] x1.27)	13.6		41.4	([x1.11] x1.15)		([x1.17] x1.19)
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	2.15		0.00120	x2.18		4.55	2.15		0.00981	x2.44		5.08	2.41		0.0872	x2.78		9.47
		run [μ s/pt]	0.922		1.19	([x3.54] x3.83)		([x3.06] x3.34)	2.92		6.57	([x1.87] x2.00)		([x1.24] x1.37)	9.32		42.1	([x1.14] x1.18)		([x1.16] x1.22)
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	2.16		0.000960	x1.95		4.15	2.40		0.00710	x1.97		4.38	2.81		0.0621	x1.99		5.37
		run [μ s/pt]	0.450		0.416	([x2.26] x2.60)		([x2.04] x2.47)	1.37		1.38	([x1.98] x2.13)		([x3.03] x3.31)	4.08		7.30	([x1.35] x1.44)		([x1.19] x1.28)
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	2.20		0.00102	x1.97		4.39	2.22		0.00727	x2.00		4.45	2.46		0.0800	x3.37		5.31
		run [μ s/pt]	0.162		0.360	([x2.99] x3.84)		([x2.04] x2.56)	0.626		1.23	([x4.51] x4.85)		([x2.95] x3.22)	2.23		7.06	([x2.90] x2.71)		([x1.12] x1.20)
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	2.41		0.00107	x1.99		4.29	3.53		0.00791	x1.87		4.60	5.24		0.0679	x1.72		6.51
		run [μ s/pt]	2.12		0.686	([x1.76] x1.89)		([x2.45] x2.81)	6.60		2.56	([x1.44] x1.50)		([x3.14] x3.34)	18.1		13.5	([x1.26] x1.33)		([x1.27] x1.34)
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.22		0.00198	x1.54		4.70	23.1		0.0186	x1.56		7.66	96.5		0.208	x1.29		28.7
		run [μ s/pt]	2.46		3.34	([x1.92] x2.06)		([x1.62] x1.70)	8.19		25.6	([x1.49] x1.57)		([x1.04] x1.07)	23.0		198	([x1.41] x1.45)		([x1.14] x1.15)
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	3.43		0.00123	x1.81		4.53	6.72		0.00991	x1.47		5.29	11.2		0.0958	x2.13		11.6
		run [μ s/pt]	0.895		1.56	([x3.05] x3.37)		([x3.09] x3.36)	3.03		9.64	([x1.83] x1.99)		([x1.10] x1.15)	9.31		65.5	([x1.53] x1.64)		([x1.16] x1.14)
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]	3.20		0.00393	x2.96		6.11	4.60		0.0636	x5.37		20.4	10.7		2.54	x7.12		192
		run [μ s/pt]	3.32		12.4	([x1.16] x1.24)		([x0.873] x0.934)	11.1		106	([x0.965] x0.997)		([x0.881] x0.888)	39.3		1.18e+03	([x0.747] x0.761)		([x1.37] x1.36)
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.13		0.00127	x1.09		4.38	9.65		0.00703	x2.00		4.78	23.4		0.0670	x0.545		5.62
		run [μ s/pt]	4.88		0.680	([x1.60] x1.67)		([x2.37] x2.85)	13.9		1.69	([x3.17] x3.29)		([x5.85] x6.66)	37.3		6.49	([x1.15] x1.19)		([x1.29] x1.43)
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	10.3		0.000925	x0.436		4.25	13.3		0.00690	x0.353		4.46	15.7		0.0574	x0.333		4.54
		run [μ s/pt]	0.769		0.579	([x2.87] x3.24)		([x1.81] x2.18)	1.64		0.843	([x1.90] x2.11)		([x1.99] x2.32)	3.92		1.67	([x1.55] x1.68)		([x2.27] x2.60)
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	6.62		0.000955	x0.793		4.66	15.6		0.00714	x0.486		4.45	36.8		0.0632	x0.459		5.30
		run [μ s/pt]	1.11		0.717	([x2.95] x3.29)		([x2.09] x2.51)	3.29		1.91	([x2.55] x2.72)		([x2.42] x2.66)	10.5		8.25	([x2.14] x2.25)		([x1.24] x1.36)
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	11.2		0.000994	x0.435		4.65	17.2		0.00704	x0.315		4.81	17.8		0.0571	x0.310		4.48
		run [μ s/pt]	1.50		0.866	([x1.71] x1.90)		([x2.77] x3.30)	3.06		1.15	([x1.19] x1.29)		([x2.70] x2.99)	6.69		2.14	([x0.983] x1.06)		([x2.73] x2.88)
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	2.93		0.00104	x1.80		4.58	6.20		0.00749	x1.10		5.13	13.0		0.0626	x2.02		8.75
		run [μ s/pt]	0.171		0.874	([x2.77] x4.51)		([x3.06] x3.40)	0.435		3.68	([x2.42] x3.98)		([x1.52] x1.66)	1.15		24.2	([x2.26] x3.60)		([x1.28] x1.32)
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					STATIC N/A					9.13		5.83
		run [μ s/pt]											STATIC N/A					3.64		16.8
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	2.59		0.00100	x1.92		5.69	4.89		0.00784	x1.51		6.02	8.98		0.0659	x2.53		9.33
		run [μ s/pt]	0.268		1.33	([x3.58] x5.69)		([x2.04] x2.27)	0.600		6.17	([x2.10] x3.60)		([x1.59] x1.68)	1.40		36.2	([x2.03] x3.37)		([x1.62] x1.66)
summary: generation		x0.435			x2.96		x1.87		x1.66		x1.26	x0.315		x5.37		x1.71		x1.77		x1.30
summary: run single-flow, hel. sum		x1.24			x5.69		x2.92		x2.95		x2.15	x0.997		x4.85		x2.05		x2.39		x2.02
summary: run all-flows, single hel.		x0.934			x3.44		x2.69		x2.65		x1.84	x0.888		x6.66		x2.00		x2.40		x1.17

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time. Its setup boundary differs from the reference, so no generation ratio is formed. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM recurrence versus AmpliCol LC (block 3 of 3)

ID	base process	metric		n=7				n=8				n=9			
1	$dd \rightarrow Z + (n-1)g$	gen. [s]	4.68	1.08	x3.23	59.2	8.94	13.3	N/A	N/A	26.1	>15GB	N/A	N/A	
		run [μ s/pt]	38.9	323	([x0.991] x1.01)	([x1.52] x1.51)	95.8	6.18e+03	N/A	N/A	228	>15GB	N/A	N/A	
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	3.08	1.00	x3.80	58.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	27.5	310	([x0.972] x0.997)	([x1.55] x1.56)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	3.35	0.631	x2.16	10.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	12.1	45.3	([x1.11] x1.16)	([x1.15] x1.18)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	2.72	0.643	x2.34	10.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	8.99	43.4	([x0.961] x0.997)	([x1.13] x1.16)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	11.3	0.715	x1.39	19.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	47.3	84.7	([x1.16] x1.18)	([x1.29] x1.31)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	645	4.39	x0.717	373	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	61.4	3.11e+03	([x1.42] x1.45)	([x1.44] x1.46)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	36.0	1.08	x1.87	91.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	27.0	509	([x1.41] x1.47)	([x1.94] x1.99)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	30.5	209	x13.7	2.88e+03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	86.1	2.85e+04	([x0.847] x0.884)	([x1.47] x1.49)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	58.0	0.622	x0.378	12.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	83.8	28.9	([x1.28] x1.28)	([x1.20] x1.26)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	22.4	0.572	x0.287	4.93	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	9.89	5.34	([x1.36] x1.41)	([x1.28] x1.44)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	120	0.617	x0.380	9.14	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	29.7	44.9	([x1.96] x2.04)	([x1.19] x1.23)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	25.0	0.610	x0.259	5.18	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	16.1	6.26	([x0.820] x0.868)	([x1.33] x1.46)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	29.5	0.656	x3.29	32.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	2.95	142	([x2.57] x3.70)	([x1.98] x2.00)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	STATIC	N/A	STATIC	N/A	55.9	15.2	STATIC	N/A	STATIC	N/A	STATIC	N/A	
		run [μ s/pt]	STATIC	N/A	STATIC	N/A	7.17	113	STATIC	N/A	STATIC	N/A	STATIC	N/A	
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	18.5	0.684	x5.98	82.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		run [μ s/pt]	3.56	402	([x3.30] x4.77)	([x2.60] x2.61)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
summary: generation		x0.259		x13.7	x2.01	x2.84	x1.28	n=8		N/A		n=9		N/A	
summary: run single-flow, hel. sum		x0.868		x4.77	x1.23	x1.66	x1.26	N/A		N/A		N/A		N/A	
summary: run all-flows, single hel.		x1.16		x2.61	x1.46	x1.55	x1.51	N/A		N/A		>15GB			

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time. Its setup boundary differs from the reference, so no generation ratio is formed. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.5 Built-in SM recurrence versus AmpliCol NLC

Built-in SM recurrence versus AmpliCol NLC (block 1 of 2)

ID	base process	metric	n=1					n=2					n=3				
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	1.94	x2.24				1.23	x3.65				1.96	x2.34			
		run [μ s/pt]	0.0837	([x1.59] x1.98)				0.245	([x1.70] x2.00)				1.35	([x1.07] x1.25)			
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	1.93	x2.42				1.97	x2.14				2.02	x2.36			
		run [μ s/pt]	0.0520	([x1.62] x2.19)				0.134	([x1.75] x2.21)				0.763	([x1.43] x1.65)			
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]		NOT APPLICABLE				2.02	x2.10				2.13	x1.98			
		run [μ s/pt]						0.178	([x1.28] x1.52)				0.404	([x0.919] x1.10)			
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]		NOT APPLICABLE				2.09	x2.13				2.10	x2.17			
		run [μ s/pt]						0.0974	([x0.815] x1.20)				0.189	([x0.715] x1.05)			
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]		NOT APPLICABLE				2.05	x2.07				2.18	x2.00			
		run [μ s/pt]						0.334	([x1.58] x1.80)				0.992	([x1.92] x2.07)			
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				2.20	x2.01				3.23	x1.68			
		run [μ s/pt]						0.680	([x2.33] x2.61)				6.79	([x1.02] x1.19)			
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				2.07	x2.08				3.51	x1.38			
		run [μ s/pt]						0.492	([x0.795] x0.974)				2.70	([x0.697] x0.830)			
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]		NOT APPLICABLE				2.78	x1.85				2.87	x2.02			
		run [μ s/pt]						3.79	([x2.35] x2.65)				48.0	([x0.572] x0.705)			
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE				3.59	x1.31			
		run [μ s/pt]											1.95	([x1.71] x1.81)			
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
summary: generation			n=1					n=2					n=3				
summary: wall			x2.24	x2.42	x2.33	x2.33	x2.33	x1.85	x3.65	x2.09	x2.25	x2.16	x1.31	x2.36	x2.00	x1.92	x1.84
			x1.98	x2.19	x2.09	x2.09	x2.06	x0.974	x2.65	x1.90	x1.87	x2.36	x0.705	x2.07	x1.19	x1.29	x0.844

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM recurrence versus AmpliCol NLC (block 2 of 2)

ID	base process	metric	n=4					n=5					n=6				
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.09	x3.25				2.27	x12.0				NOT APPLICABLE				
		run [μ s/pt]	11.9	([x0.867] x1.02)				145	([x0.774] x0.867)								
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	2.06	x2.74				2.12	x8.44				NOT APPLICABLE				
		run [μ s/pt]	7.03	([x0.853] x0.975)				90.7	([x0.660] x0.744)								
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	2.35	x1.91				2.57	x2.42				NOT APPLICABLE				
		run [μ s/pt]	2.08	([x0.790] x0.891)				16.7	([x0.536] x0.618)								
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	2.15	x2.00				2.53	x2.00				NOT APPLICABLE				
		run [μ s/pt]	0.964	([x0.718] x0.842)				8.52	([x0.600] x0.663)								
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	2.93	x1.95				3.21	x4.26				NOT APPLICABLE				
		run [μ s/pt]	5.65	([x1.16] x1.27)				48.8	([x0.983] x1.10)								
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.25	x2.19				23.0	x6.33				NOT APPLICABLE				
		run [μ s/pt]	92.0	([x0.846] x0.942)				1.49e+03	([x1.88] x2.16)								
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	5.04	x1.41				9.64	x3.23				NOT APPLICABLE				
		run [μ s/pt]	24.8	([x0.596] x0.666)				309	([x0.711] x0.815)								
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]	4.08	x2.63				9.11	x3.14				NOT APPLICABLE				
		run [μ s/pt]	801	([x0.516] x0.645)				1.53e+04	([x0.547] x0.657)								
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.26	x1.12				21.4	x0.504				NOT APPLICABLE				
		run [μ s/pt]	5.69	([x1.38] x1.45)				44.3	([x1.14] x1.17)								
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	13.1	x0.355				14.2	x0.326				NOT APPLICABLE				
		run [μ s/pt]	1.45	([x1.53] x1.68)				3.19	([x0.998] x1.09)								
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	8.12	x0.654				20.7	x0.344				NOT APPLICABLE				
		run [μ s/pt]	4.53	([x1.18] x1.26)				26.5	([x1.06] x1.10)								
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	12.3	4.52				17.5	5.46				NOT APPLICABLE				
		run [μ s/pt]	2.90	2.68				6.22	3.72								
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	0.00110	5.14				0.0115	7.32			0.102	x407				
		run [μ s/pt]	57.2	2.07				501	15.5			6.18e+03	([x0.0290] x0.0326)				
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]		NOT APPLICABLE				NOT APPLICABLE				STATIC N/A	17.6				
		run [μ s/pt]										STATIC N/A	43.7				
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	0.00106	5.07				0.00879	x1.17e+03			0.103	x688				
		run [μ s/pt]	84.1	3.30				820	([x0.0280] x0.0334)			9.90e+03	([x0.0467] x0.0509)				
summary: generation			x0.355	x3.25	x1.95	x1.84	x1.40	x0.326	x1.17e+03	x3.19	x101	x2.78	x407	x688	x547	x547	x548
summary: wall			x0.645	x1.68	x0.975	x1.06	x0.695	x0.0334	x2.16	x0.841	x0.918	x0.759	x0.0326	x0.0509	x0.0417	x0.0417	x0.0439

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.6 Built-in SM recurrence versus AmpliCol full-colour

Built-in SM recurrence versus AmpliCol full-colour (block 1 of 2)

ID	base process	metric	n=1					n=2					n=3				
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.25	x1.99				1.94	x2.25				2.21	x2.30			
		run [μ s/pt]	0.0993	([x1.41] x1.72)				0.275	([x1.45] x1.67)				1.60	([x0.923] x1.08)			
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	2.00	x2.16				2.04	x2.14				1.96	x2.39			
		run [μ s/pt]	0.0595	([x1.38] x1.83)				0.152	([x1.49] x1.86)				0.901	([x1.35] x1.57)			
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]		NOT APPLICABLE				2.18	4.49				2.28	4.52			
		run [μ s/pt]						0.211	0.282				0.457	0.469			
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]		NOT APPLICABLE				2.10	x2.14				2.14	x2.10			
		run [μ s/pt]						0.107	([x0.741] x1.09)				0.209	([x0.703] x0.990)			
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]		NOT APPLICABLE				2.10	x2.02				2.65	x1.86			
		run [μ s/pt]						0.375	([x1.36] x1.59)				1.14	([x1.71] x1.86)			
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				2.12	x2.19				3.93	x1.44			
		run [μ s/pt]						0.831	([x2.21] x2.43)				8.55	([x0.815] x0.996)			
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				2.06	x2.09				3.57	x1.33			
		run [μ s/pt]						0.622	([x0.646] x0.813)				3.19	([x0.608] x0.717)			
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]		NOT APPLICABLE				4.63	x1.81				2.88	x2.03			
		run [μ s/pt]						5.93	([x1.80] x2.06)				61.4	([x0.460] x0.561)			
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE				4.10	x1.18			
		run [μ s/pt]											2.09	([x1.65] x1.78)			
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
summary: generation			n=1					n=2					n=3				
summary: wall			x1.99	x2.16	x2.07	x2.07	x2.07	x1.81	x2.25	x2.14	x2.09	x2.05	x1.18	x2.39	x1.95	x1.83	x1.72
			x1.72	x1.83	x1.78	x1.78	x1.76	x0.813	x2.43	x1.67	x1.64	x1.95	x0.561	x1.86	x1.04	x1.20	x0.689

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM recurrence versus AmpliCol full-colour (block 2 of 2)

ID	base process	metric		n=4		n=5		n=6										
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.07	x3.29		2.49	x11.3		NOT APPLICABLE									
		run [μ s/pt]	14.2	([x0.715] x0.872)		203	([x0.551] x0.656)											
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	2.15	x2.73		2.16	x8.19		NOT APPLICABLE									
		run [μ s/pt]	8.30	([x0.721] x0.874)		116	([x0.514] x0.586)											
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	2.46	4.75		2.55	x2.45		NOT APPLICABLE									
		run [μ s/pt]	2.40	1.93		20.1	([x0.450] x0.531)											
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	2.18	x2.08		2.36	x2.10		NOT APPLICABLE									
		run [μ s/pt]	1.12	([x0.594] x0.719)		10.2	([x0.512] x0.569)											
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	2.51	x2.25		3.24	x4.46		NOT APPLICABLE									
		run [μ s/pt]	6.06	([x1.08] x1.22)		55.4	([x0.883] x1.01)											
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.27	x2.19		21.6	x6.75		NOT APPLICABLE									
		run [μ s/pt]	126	([x0.608] x0.687)		2.82e+03	([x0.962] x1.17)											
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	4.56	x1.59		11.2	x2.93		NOT APPLICABLE									
		run [μ s/pt]	32.9	([x0.448] x0.501)		555	([x0.401] x0.474)											
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	4.11	x2.62		7.79	x4.22		NOT APPLICABLE									
		run [μ s/pt]	1.38e+03	([x0.303] x0.397)		5.03e+04	([x0.156] x0.240)											
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.02	x1.13		9.64	x1.09		NOT APPLICABLE									
		run [μ s/pt]	5.97	([x1.45] x1.46)		32.7	([x1.04] x1.12)											
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	11.4	x0.402		14.3	x0.337		NOT APPLICABLE									
		run [μ s/pt]	1.50	([x1.43] x1.57)		3.30	([x0.947] x1.03)											
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	7.81	x0.625		20.3	x0.345		NOT APPLICABLE									
		run [μ s/pt]	5.00	([x1.07] x1.13)		28.6	([x0.970] x1.01)											
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	12.0	4.52		16.1	4.71		NOT APPLICABLE									
		run [μ s/pt]	2.93	2.53		6.29	3.59											
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	0.00106	4.91		0.00894	7.83	0.120	x437									
		run [μ s/pt]	59.5	2.06		536	16.0	7.98e+03	([x0.0350] x0.0410)									
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE			NOT APPLICABLE			STATIC N/A	17.7								
		run [μ s/pt]							STATIC N/A	45.3								
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	0.00147	4.94		0.00866	10.2	0.106	x675									
		run [μ s/pt]	86.2	3.22		842	27.0	1.06e+04	([x0.0442] x0.0482)									
summary: generation				n=4				n=5				n=6						
				x0.402	x3.29	x2.13	x1.89	x1.47	x0.337	x11.3	x2.93	x4.01	x3.13	x437	x675	x556	x556	x549
summary: wall				x0.397	x1.57	x0.873	x0.942	x0.440	x0.240	x1.17	x0.656	x0.763	x0.295	x0.0410	x0.0482	x0.0446	x0.0446	x0.0451

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.7 UFO-SM recurrence versus AmpliCol LC

UFO-SM recurrence versus AmpliCol LC (block 1 of 3)

ID	base process	metric	n=1					n=2					n=3				
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.14	5.50e-05	x3.49	7.78	1.96	8.50e-05	x3.84	7.36	2.19	0.000263	x3.51	7.64			
		run [μ s/pt]	0.0218	0.0473	([x7.38] x9.22)	([x1.12] x2.35)	0.127	0.111	([x3.58] x4.25)	([x1.58] x2.43)	0.461	0.328	([x2.58] x2.89)	([x2.12] x2.58)			
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	1.93	5.60e-05	x3.81	7.78	1.95	8.10e-05	x4.84	7.63	2.07	0.000260	x3.58	7.31			
		run [μ s/pt]	0.0133	0.0467	([x5.92] x8.93)	([x0.683] x2.11)	0.0734	0.115	([x3.35] x4.19)	([x1.32] x2.13)	0.267	0.317	([x2.52] x2.99)	([x1.94] x2.41)			
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE					2.16	7.60e-05	x3.50	7.69	2.26	0.000222	x3.46	7.56		
		run [μ s/pt]						0.0691	0.118	([x3.61] x4.58)	([x1.46] x2.17)	0.177	0.196	([x2.37] x2.91)	([x1.57] x2.16)		
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE					2.06	7.40e-05	x3.77	7.68	2.09	0.000218	x3.67	7.74		
		run [μ s/pt]						0.0331	0.0899	([x2.44] x4.19)	([x0.768] x1.71)	0.0638	0.160	([x2.42] x3.80)	([x1.22] x1.98)		
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE					2.05	7.50e-05	x3.61	7.79	2.16	0.000227	x3.54	7.40		
		run [μ s/pt]						0.159	0.115	([x4.95] x5.59)	([x1.65] x2.51)	0.718	0.243	([x3.39] x3.58)	([x2.04] x2.63)		
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE					2.17	8.70e-05	x3.47	7.52	3.24	0.000340	x2.68	7.39		
		run [μ s/pt]						0.102	0.155	([x9.65] x10.5)	([x2.10] x2.80)	0.704	0.639	([x3.13] x3.40)	([x3.05] x3.35)		
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE					2.02	8.30e-05	x3.68	7.24	2.51	0.000257	x3.24	7.65		
		run [μ s/pt]						0.0501	0.112	([x4.92] x6.17)	([x1.15] x1.99)	0.242	0.379	([x4.65] x5.24)	([x2.06] x2.51)		
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	NOT APPLICABLE					2.85	0.000112	x2.82	7.64	2.66	0.000563	x3.19	8.71		
		run [μ s/pt]						0.187	0.250	([x7.83] x8.59)	([x5.53] x6.11)	0.970	1.54	([x2.85] x2.89)	([x1.97] x2.13)		
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					3.60	0.000246	x2.23	7.68	
		run [μ s/pt]											1.54	0.390	([x2.78] x2.92)	([x1.98] x2.51)	
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE				
		run [μ s/pt]															
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE				
		run [μ s/pt]															
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE				
		run [μ s/pt]															
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE				
		run [μ s/pt]															
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE				
		run [μ s/pt]															
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE				
		run [μ s/pt]															
summary: generation			n=1					n=2					n=3				
summary: run single-flow, hel. sum			x3.49	x3.81	x3.65	x3.65	x3.64	x2.82	x4.84	x3.65	x3.69	x3.64	x2.23	x3.67	x3.46	x3.23	x3.14
summary: run all-flows, single hel.			x8.93	x9.22	x9.07	x9.07	x9.11	x4.19	x10.5	x5.09	x6.01	x6.47	x2.89	x5.24	x2.99	x3.40	x3.19
			x2.11	x2.35	x2.23	x2.23	x2.23	x1.71	x6.11	x2.30	x2.73	x3.19	x1.98	x3.35	x2.51	x2.47	x2.47

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time. Its setup boundary differs from the reference, so no generation ratio is formed. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

UFO-SM recurrence versus AmpliCol LC (block 2 of 3)

ID	base process	metric		n=4				n=5				n=6																
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.08		0.00124	$x3.83$		7.54	2.34		0.00975	$x3.76$		8.10	2.95		0.0884	$x3.93$		13.2								
		run [μ s/pt]	1.46		1.19	([x2.19] $x2.36$)		([x2.96] $x3.20$)	4.56		6.53	([x1.44] $x1.52$)		([x1.22] $x1.31$)	13.6		41.4	([x1.18] $x1.22$)		([x1.21] $x1.24$)								
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	2.15		0.00120	$x3.63$		7.98	2.15		0.00981	$x4.14$		8.45	2.41		0.0872	$x4.13$		12.8								
		run [μ s/pt]	0.922		1.19	([x3.81] $x3.98$)		([x2.84] $x3.07$)	2.92		6.57	([x2.06] $x2.15$)		([x1.18] $x1.27$)	9.32		42.1	([x1.10] $x1.15$)		([x1.21] $x1.20$)								
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	2.16		0.000960	$x3.67$		7.57	2.40		0.00710	$x3.24$		7.38	2.81		0.0621	$x3.06$		8.07								
		run [μ s/pt]	0.450		0.416	([x2.75] $x3.03$)		([x2.28] $x2.67$)	1.37		1.38	([x2.08] $x2.29$)		([x2.67] $x2.90$)	4.08		7.30	([x1.41] $x1.50$)		([x1.21] $x1.29$)								
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	2.20		0.00102	$x3.33$		7.17	2.22		0.00727	$x3.41$		7.62	2.46		0.0800	$x3.38$		8.47								
		run [μ s/pt]	0.162		0.360	([x2.94] $x3.89$)		([x1.88] $x2.36$)	0.626		1.23	([x4.64] $x5.03$)		([x2.62] $x2.91$)	2.23		7.06	([x2.09] $x2.27$)		([x1.10] $x1.19$)								
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	2.41		0.00107	$x3.38$		7.31	3.53		0.00791	$x2.68$		7.87	5.24		0.0679	$x2.97$		9.80								
		run [μ s/pt]	2.12		0.686	([x2.16] $x2.27$)		([x2.62] $x2.93$)	6.60		2.56	([x1.63] $x1.70$)		([x3.09] $x3.31$)	18.1		13.5	([x1.45] $x1.49$)		([x1.32] $x1.40$)								
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.22		0.00198	$x2.15$		7.79	23.1		0.0186	$x1.83$		11.4	96.5		0.208	$x1.43$		33.9								
		run [μ s/pt]	2.46		3.34	([x2.00] $x2.11$)		([x1.59] $x1.70$)	8.19		25.6	([x1.52] $x1.60$)		([x1.07] $x1.11$)	23.0		198	([x1.45] $x1.50$)		([x1.18] $x1.18$)								
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	3.43		0.00123	$x2.69$		7.45	6.72		0.00991	$x2.25$		8.70	11.2		0.0958	$x2.70$		15.5								
		run [μ s/pt]	0.895		1.56	([x2.87] $x3.15$)		([x3.06] $x3.24$)	3.03		9.64	([x1.86] $x2.05$)		([x1.06] $x1.10$)	9.31		65.5	([x1.58] $x1.69$)		([x1.12] $x1.14$)								
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	3.20		0.00393	$x4.05$		10.0	4.60		0.0636	$x6.38$		28.4	10.7		2.54	$x7.66$		213								
		run [μ s/pt]	3.32		12.4	([x1.16] $x1.25$)		([x0.913] $x0.956$)	11.1		106	([x1.00] $x1.07$)		([x0.960] $x0.976$)	39.3		1.18e+03	([x0.721] $x0.752$)		([x1.31] $x1.31$)								
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.13		0.00127	$x1.74$		7.38	9.65		0.00703	$x2.61$		8.95	23.4		0.0670	$x0.693$		8.83								
		run [μ s/pt]	4.88		0.680	([x2.12] $x2.19$)		([x2.14] $x2.53$)	13.9		1.69	([x2.98] $x3.20$)		([x7.09] $x7.85$)	37.3		6.49	([x1.56] $x1.57$)		([x1.35] $x1.49$)								
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	10.3		0.000925	$x0.735$		7.22	13.3		0.00690	$x0.588$		7.32	15.7		0.0574	$x0.548$		7.47								
		run [μ s/pt]	0.769		0.579	([x3.25] $x3.54$)		([x1.80] $x2.18$)	1.64		0.843	([x2.35] $x2.55$)		([x2.01] $x2.38$)	3.92		1.67	([x1.94] $x2.06$)		([x2.28] $x2.56$)								
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	6.62		0.000955	$x1.25$		7.25	15.6		0.00714	$x0.721$		7.77	36.8		0.0632	$x0.588$		8.53								
		run [μ s/pt]	1.11		0.717	([x3.10] $x3.35$)		([x1.90] $x2.23$)	3.29		1.91	([x2.63] $x2.80$)		([x2.48] $x2.64$)	10.5		8.25	([x2.23] $x2.34$)		([x1.29] $x1.40$)								
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	11.2		0.000994	$x0.677$		7.44	17.2		0.00704	$x0.459$		7.35	17.8		0.0571	$x0.490$		7.68								
		run [μ s/pt]	1.50		0.866	([x1.68] $x1.87$)		([x2.10] $x2.30$)	3.06		1.15	([x1.26] $x1.37$)		3.00	6.69		2.14	([x1.18] $x1.25$)		([x2.25] $x2.47$)								
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	2.93		0.00104	$x2.82$		9.54	6.20		0.00749	10.4		8.06	13.0		0.0626	$x2.34$		10.6								
		run [μ s/pt]	0.171		0.874	([x2.58] $x4.22$)		([x4.27] $x4.48$)	0.435		3.68	1.60		6.19	1.15		24.2	([x2.14] $x3.44$)		([x1.32] $x1.36$)								
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				STATIC N/A		STATIC N/A		12.9		9.30											
		run [μ s/pt]									STATIC N/A		STATIC N/A		3.78		18.7											
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	2.59		0.00100	$x2.89$		7.40	4.89		0.00784	9.77		8.38	8.98		0.0659	$x3.22$		13.1								
		run [μ s/pt]	0.268		1.33	([x2.11] $x3.49$)		2.56	0.600		6.17	2.31		10.2	1.40		36.2	([x2.01] $x3.37$)		([x1.73] $x1.79$)								
summary: generation		$x0.677$		$x4.05$		$x2.86$		$x2.63$		$x1.97$	$x0.459$		$x6.38$		$x2.64$		$x2.67$		$x1.76$	$x0.490$		$x7.66$		$x2.83$		$x2.65$		$x1.67$
summary: run single-flow, hel. sum		$x1.25$		$x4.22$		$x3.09$		$x2.91$		$x2.33$	$x1.07$		$x5.03$		$x2.10$		$x2.28$		$x2.06$	$x0.752$		$x3.44$		$x1.54$		$x1.83$		$x1.41$
summary: run all-flows, single hel.		$x0.956$		$x4.48$		$x2.53$		$x2.60$		$x1.79$	$x0.976$		$x7.85$		$x2.38$		$x2.52$		$x1.19$	$x1.14$		$x2.56$		$x1.34$		$x1.50$		$x1.30$

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time. Its setup boundary differs from the reference, so no generation ratio is formed. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

UFO-SM recurrence versus AmpliCol LC (block 3 of 3)

ID	base process	metric		n=7				n=8				n=9								
1	$dd \rightarrow Z + (n-1)g$	gen. [s]	4.68		1.08	x4.23		66.3	8.94		13.3	N/A		N/A	26.1		>15GB	N/A		N/A
		run [μ s/pt]	38.9		323	([x1.07] x1.08)		([x1.62] x1.62)	95.8		6.18e+03	N/A		N/A	228		>15GB	N/A		N/A
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	3.08		1.00	x5.01		68.9	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	27.5		310	([x0.980] x1.04)		([x1.64] x1.65)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
3	$dd \rightarrow e^+e^- + (n-2)g$	gen. [s]	3.35		0.631	x3.20		13.9	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	12.1		45.3	([x1.17] x1.23)		([x1.18] x1.21)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	2.72		0.643	x3.58		13.7	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	8.99		43.4	([x0.930] x0.983)		([x1.14] x1.18)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
5	$dd \rightarrow ZZ + (n-2)g$	gen. [s]	11.3		0.715	x1.75		23.7	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	47.3		84.7	([x1.33] x1.36)		([x1.27] x1.29)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	645		4.39	x0.776		397	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	61.4		3.11e+03	([x1.43] x1.47)		([x1.49] x1.48)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
7	$dd \rightarrow t\bar{t} + (n-2)g$	gen. [s]	36.0		1.08	x2.24		100	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	27.0		509	([x1.50] x1.59)		([x1.97] x1.98)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	30.5		209	x13.8		2.91e+03	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	86.1		2.85e+04	([x0.855] x0.895)		([x1.47] x1.47)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
9	$dd \rightarrow ZZZ + (n-3)g$	gen. [s]	58.0		0.622	x0.475		15.7	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	83.8		28.9	([x1.57] x1.59)		([x1.28] x1.32)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
10	$dd \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	22.4		0.572	x0.445		8.02	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	9.89		5.34	([x1.66] x1.73)		([x1.41] x1.57)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
11	$dd \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	120		0.617	x0.453		12.7	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	29.7		44.9	([x2.15] x2.20)		([x1.24] x1.30)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
12	$dd \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	25.0		0.610	x0.396		8.48	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	16.1		6.26	([x0.968] x1.02)		([x1.32] x1.49)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
13	$dd \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	29.5		0.656	x3.81		37.5	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	2.95		142	([x2.74] x3.78)		([x2.15] x2.17)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
14	$dd \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	STATIC	N/A		STATIC	N/A	65.2		19.8	STATIC	N/A		STATIC	N/A		STATIC	N/A		N/A
		run [μ s/pt]	STATIC	N/A		STATIC	N/A	7.26		117	STATIC	N/A		STATIC	N/A		STATIC	N/A		N/A
15	$dd \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	18.5		0.684	x7.33		95.7	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
		run [μ s/pt]	3.56		402	([x4.15] x5.83)		([x2.02] x2.00)	N/A		N/A	N/A		N/A	N/A		N/A	N/A		N/A
summary: generation		x0.396			x13.8		x2.72		x3.39		x1.41	n=8		N/A		n=9		N/A		
summary: run single-flow, hel. sum		x0.895			x5.83		x1.42		x1.84		x1.39	N/A		N/A		N/A		N/A		
summary: run all-flows, single hel.		x1.18			x2.17		x1.48		x1.55		x1.49	N/A		>15GB						

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the selected-flow library-generation and all-flow direct-setup baseline quantities. The selected-flow entry carries a generation ratio; the all-flow entry carries the absolute pyAmpliCol process-generation time. Its setup boundary differs from the reference, so no generation ratio is formed. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.8 UFO-SM recurrence versus AmpliCol NLC

UFO-SM recurrence versus AmpliCol NLC (block 1 of 2)

ID	base process	metric	n=1					n=2					n=3				
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	1.94	x3.75				1.23	x6.21				1.96	x4.00			
		run [μ s/pt]	0.0837	([x1.41] x1.83)				0.245	([x1.88] x2.17)				1.35	([x1.22] x1.39)			
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	1.93	x4.34				1.97	x4.06				2.02	x3.77			
		run [μ s/pt]	0.0520	([x1.58] x2.14)				0.134	([x1.72] x2.15)				0.763	([x1.32] x1.56)			
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]		NOT APPLICABLE				2.02	x3.85				2.13	x3.84			
		run [μ s/pt]						0.178	([x1.38] x1.62)				0.404	([x1.07] x1.27)			
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]		NOT APPLICABLE				2.09	x3.71				2.10	x3.66			
		run [μ s/pt]						0.0974	([x0.812] x1.20)				0.189	([x0.748] x1.08)			
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]		NOT APPLICABLE				2.05	x3.67				2.18	x3.44			
		run [μ s/pt]						0.334	([x1.73] x1.99)				0.992	([x2.28] x2.44)			
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				2.20	x3.56				3.23	x2.75			
		run [μ s/pt]						0.680	([x2.56] x2.79)				6.79	([x1.06] x1.22)			
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				2.07	x3.51				3.51	x2.21			
		run [μ s/pt]						0.492	([x0.901] x1.09)				2.70	([x0.709] x0.834)			
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]		NOT APPLICABLE				2.78	x2.82				2.87	x3.10			
		run [μ s/pt]						3.79	([x2.39] x2.66)				48.0	([x0.583] x0.714)			
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE				3.59	x2.19			
		run [μ s/pt]											1.95	([x2.14] x2.27)			
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
summary: generation			n=1					n=2					n=3				
summary: wall			x3.75	x4.34	x4.04	x4.04	x4.04	x2.82	x6.21	x3.69	x3.92	x3.76	x2.19	x4.00	x3.44	x3.22	x3.06
			x1.83	x2.14	x1.99	x1.99	x1.95	x1.09	x2.79	x2.07	x1.96	x2.42	x0.714	x2.44	x1.27	x1.42	x0.878

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

UFO-SM recurrence versus AmpliCol NLC (block 2 of 2)

ID	base process	metric		n=4			n=5			n=6						
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.09	x4.92			2.27	x15.7			NOT APPLICABLE					
		run [μ s/pt]	11.9	([x0.959] x1.12)			145	([x0.840] x0.945)								
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	2.06	x4.40			2.12	x10.9			NOT APPLICABLE					
		run [μ s/pt]	7.03	([x0.839] x0.970)			90.7	([x0.657] x0.742)								
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	2.35	x3.31			2.57	x3.77			NOT APPLICABLE					
		run [μ s/pt]	2.08	([x0.863] x0.952)			16.7	([x0.581] x0.652)								
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	2.15	x3.92			2.53	x3.26			NOT APPLICABLE					
		run [μ s/pt]	0.964	([x0.715] x0.882)			8.52	([x0.611] x0.668)								
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	2.93	x3.10			3.21	x5.92			NOT APPLICABLE					
		run [μ s/pt]	5.65	([x1.41] x1.53)			48.8	([x1.24] x1.32)								
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.25	x2.99			23.0	x7.73			NOT APPLICABLE					
		run [μ s/pt]	92.0	([x0.863] x0.969)			1.49e+03	([x1.94] x2.22)								
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	5.04	x2.16			9.64	x4.10			NOT APPLICABLE					
		run [μ s/pt]	24.8	([x0.619] x0.693)			309	([x0.744] x0.838)								
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	4.08	x3.77			9.11	x3.80			NOT APPLICABLE					
		run [μ s/pt]	801	([x0.519] x0.666)			1.53e+04	([x0.498] x0.629)								
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.26	x1.70			21.4	x1.19			NOT APPLICABLE					
		run [μ s/pt]	5.69	([x1.80] x1.89)			44.3	([x2.43] x2.51)								
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	13.1	x0.565			14.2	x0.574			NOT APPLICABLE					
		run [μ s/pt]	1.45	([x1.67] x1.82)			3.19	([x1.19] x1.27)								
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	8.12	x0.968			20.7	x0.511			NOT APPLICABLE					
		run [μ s/pt]	4.53	([x1.17] x1.24)			26.5	([x1.10] x1.14)								
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	12.3	7.68			17.5	8.41			NOT APPLICABLE					
		run [μ s/pt]	2.90	2.73			6.22	4.42								
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	0.00110	x7.25e+03			0.0115	x963			0.102	x527				
		run [μ s/pt]	57.2	([x0.0294] x0.0366)			501	([x0.0258] x0.0320)			6.18e+03	([x0.0301] x0.0336)				
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE			NOT APPLICABLE			STATIC N/A		22.8					
		run [μ s/pt]							STATIC N/A		44.9					
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	0.00106	8.20			0.00879	x1.63e+03			0.103	x822				
		run [μ s/pt]	84.1	3.40			820	([x0.0283] x0.0334)			9.90e+03	([x0.0510] x0.0556)				
summary: generation summary: wall		n=4					n=5					n=6				
		x0.565	x7.25e+03	x3.20	x607	x2.29	x0.511	x1.63e+03	x4.10	x204	x3.77	x527	x822	x674	x674	x676
		x0.0366	x1.89	x0.969	x1.06	x0.683	x0.0320	x2.51	x0.838	x1.00	x0.727	x0.0336	x0.0556	x0.0446	x0.0446	x0.0472

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.9 UFO-SM recurrence versus AmpliCol full-colour

UFO-SM recurrence versus AmpliCol full-colour (block 1 of 2)

ID	base process	metric	n=1					n=2					n=3				
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.25	x3.30				1.94	x3.85				2.21	x3.62			
		run [μ s/pt]	0.0993	([x1.22] x1.58)				0.275	([x1.69] x1.96)				1.60	([x1.09] x1.25)			
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	2.00	x3.69				2.04	x3.74				1.96	x3.93			
		run [μ s/pt]	0.0595	([x1.27] x1.76)				0.152	([x1.57] x1.92)				0.901	([x1.11] x1.31)			
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]		NOT APPLICABLE				2.18	7.99				2.28	7.68			
		run [μ s/pt]						0.211	0.287				0.457	0.537			
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]		NOT APPLICABLE				2.10	x3.63				2.14	x3.73			
		run [μ s/pt]						0.107	([x0.745] x1.06)				0.209	([x0.736] x1.00)			
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]		NOT APPLICABLE				2.10	x3.86				2.65	x2.92			
		run [μ s/pt]						0.375	([x2.05] x2.35)				1.14	([x2.09] x2.28)			
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				2.12	x3.76				3.93	x2.30			
		run [μ s/pt]						0.831	([x2.28] x2.49)				8.55	([x0.968] x1.22)			
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				2.06	x3.54				3.57	x2.22			
		run [μ s/pt]						0.622	([x0.676] x0.812)				3.19	([x0.610] x0.720)			
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]		NOT APPLICABLE				4.63	x2.05				2.88	x3.21			
		run [μ s/pt]						5.93	([x1.63] x1.74)				61.4	([x0.484] x0.598)			
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE				4.10	x2.05			
		run [μ s/pt]											2.09	([x2.05] x2.21)			
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
summary: generation			n=1					n=2					n=3				
summary: wall			x3.30	x3.69	x3.50	x3.50	x3.49	x2.05	x3.86	x3.74	x3.49	x3.27	x2.05	x3.93	x3.06	x3.00	x2.82
			x1.58	x1.76	x1.67	x1.67	x1.65	x0.812	x2.49	x1.92	x1.76	x1.78	x0.598	x2.28	x1.23	x1.32	x0.759

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

UFO-SM recurrence versus AmpliCol full-colour (block 2 of 2)

ID	base process	metric	n=4			n=5			n=6								
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	2.07	x4.98		2.49	x14.2		NOT APPLICABLE								
		run [μ s/pt]	14.2	([x0.802] x0.961)		203	([x0.614] x0.688)										
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	2.15	x4.20		2.16	x10.7		NOT APPLICABLE								
		run [μ s/pt]	8.30	([x0.773] x0.878)		116	([x0.520] x0.584)										
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	2.46	7.86		2.55	x3.72		NOT APPLICABLE								
		run [μ s/pt]	2.40	2.08		20.1	([x0.485] x0.560)										
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	2.18	x3.47		2.36	x3.44		NOT APPLICABLE								
		run [μ s/pt]	1.12	([x0.600] x0.719)		10.2	([x0.510] x0.570)										
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	2.51	x3.59		3.24	x5.88		NOT APPLICABLE								
		run [μ s/pt]	6.06	([x1.31] x1.40)		55.4	([x1.09] x1.22)										
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.27	x3.02		21.6	x8.27		NOT APPLICABLE								
		run [μ s/pt]	126	([x0.628] x0.707)		2.82e+03	([x0.974] x1.18)										
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	4.56	x2.39		11.2	x3.61		NOT APPLICABLE								
		run [μ s/pt]	32.9	([x0.464] x0.515)		555	([x0.425] x0.503)										
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	4.11	x3.47		7.79	x4.90		NOT APPLICABLE								
		run [μ s/pt]	1.38e+03	([x0.300] x0.398)		5.03e+04	([x0.207] x0.324)										
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.02	x1.80		9.64	x1.59		NOT APPLICABLE								
		run [μ s/pt]	5.97	([x1.73] x1.80)		32.7	([x2.63] x2.72)										
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	11.4	x0.691		14.3	x0.548		NOT APPLICABLE								
		run [μ s/pt]	1.50	([x1.65] x1.83)		3.30	([x1.17] x1.29)										
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	7.81	x1.03		20.3	x0.515		NOT APPLICABLE								
		run [μ s/pt]	5.00	([x1.17] x1.25)		28.6	([x1.01] x1.04)										
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	12.0	7.49		16.1	7.89		NOT APPLICABLE								
		run [μ s/pt]	2.93	2.87		6.29	4.09										
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	0.00106	8.03		0.00894	x1.30e+03		0.120	x399							
		run [μ s/pt]	59.5	2.17		536	([x0.0247] x0.0302)		7.98e+03	([x0.0261] x0.0289)							
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE			NOT APPLICABLE			STATIC N/A		22.8						
		run [μ s/pt]							STATIC N/A		44.5						
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	0.00147	8.00		0.00866	x1.82e+03		0.106	x814							
		run [μ s/pt]	86.2	3.27		842	([x0.0277] x0.0327)		1.06e+04	([x0.0471] x0.0506)							
summary: generation			n=4			n=5			n=6								
			x0.691	x4.98	x3.25	x2.87	x2.20	x0.515	x1.82e+03	x4.90	x245	x4.23	x399	x814	x606	x606	x593
summary: wall			x0.398	x1.83	x0.920	x1.05	x0.446	x0.0302	x2.72	x0.584	x0.827	x0.366	x0.0289	x0.0506	x0.0397	x0.0397	x0.0413

Baseline: original AmpliCol; candidate: recurrence JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Original AmpliCol is static N/A beyond three open quark lines; unavailable, terminal, or unlinked baselines keep their status while successful candidate timings remain absolute and leave summaries. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.10 Built-in SM compiled JIT O3 versus recurrence JIT O2 LC

Built-in SM compiled JIT O3 versus recurrence JIT O2 LC (block 1 of 3)

ID	base process	metric	n=1					n=2					n=3																
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	4.50		4.24	x0.503		x0.465	4.46		4.25	x0.546		x0.516	4.92		5.08	x0.725		x0.544									
		run [μ s/pt]	0.169		0.0924	([x1.02] x1.02)		([x0.938] x0.938)	0.470		0.258	([x0.942] x0.941)		([x0.606] x0.606)	1.30		0.843	([x0.583] x0.583)		([x0.368] x0.367)									
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	4.33		4.35	x0.520		x0.469	4.43		4.40	x0.524		x0.466	4.38		4.47	x0.673		x0.513									
		run [μ s/pt]	0.117		0.0934	([x1.12] x1.12)		([x1.01] x1.01)	0.300		0.260	([x1.08] x1.08)		([x0.609] x0.609)	0.835		0.845	([x0.607] x0.607)		([x0.368] x0.367)									
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE					4.49		4.45	x0.620		x0.570	4.45		4.41	x0.714		x0.634										
		run [μ s/pt]						0.295		0.243	([x0.812] x0.811)		([x0.663] x0.663)	0.512		0.422	([x0.845] x0.844)		([x0.543] x0.543)										
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE					4.42		4.45	x0.507		x0.465	4.50		4.48	x0.537		x0.455										
		run [μ s/pt]						0.135		0.189	([x0.955] x0.955)		([x0.749] x0.749)	0.232		0.343	([x1.06] x1.06)		([x0.606] x0.605)										
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE					4.22		4.04	x0.580		x0.612	4.38		4.23	x0.672		x0.554										
		run [μ s/pt]						0.593		0.282	([x0.818] x0.818)		([x0.564] x0.563)	2.10		0.660	([x0.567] x0.567)		([x0.413] x0.413)										
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE					4.38		4.15	x0.567		x0.513	5.39		4.43	x0.974		x0.646										
		run [μ s/pt]						1.04		0.400	([x0.376] x0.376)		([x0.502] x0.502)	2.37		2.13	([x0.484] x0.484)		([x0.308] x0.308)										
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE					4.28		4.21	x0.611		x0.535	5.17		4.83	x0.791		x0.656										
		run [μ s/pt]						0.290		0.221	([x0.898] x0.897)		([x0.723] x0.723)	1.28		1.16	([x0.461] x0.461)		([x0.316] x0.316)										
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	NOT APPLICABLE					4.78		4.55	x0.624		x0.550	5.53		4.50	x1.24		x0.759										
		run [μ s/pt]						1.52		1.53	([x0.218] x0.218)		([x0.158] x0.158)	2.99		3.65	([x0.446] x0.446)		([x0.329] x0.329)										
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					4.91		4.26	x0.752		x0.564											
		run [μ s/pt]											3.73		0.868	([x0.725] x0.724)		([x0.409] x0.409)											
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE																
		run [μ s/pt]																											
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE																
		run [μ s/pt]																											
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE																
		run [μ s/pt]																											
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE																
		run [μ s/pt]																											
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE																
		run [μ s/pt]																											
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE																
		run [μ s/pt]																											
summary: generation			x0.503		x0.520		x0.512		x0.512		x0.511	x0.507		x0.624		x0.573		x0.572		x0.573	x0.537		x1.24		x0.725		x0.787		x0.801
summary: run single-flow, hel. sum			x1.02		x1.12		x1.07		x1.07		x1.06	x0.218		x1.08		x0.858		x0.762		x0.561	x0.446		x1.06		x0.583		x0.641		x0.580
summary: run all-flows, single hel.			x0.938		x1.01		x0.973		x0.973		x0.973	x0.158		x0.749		x0.607		x0.572		x0.408	x0.308		x0.605		x0.367		x0.406		x0.358

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. Runtime comparisons use (xS)xW. The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM compiled JIT O3 versus recurrence JIT O2 LC (block 2 of 3)

ID	base process	metric	n=4						n=5						n=6														
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	4.82		4.33		x1.30		x1.11	5.50		4.95		x3.48		x3.21	7.76		9.39		x14.1		x10.6						
		run [μ s/pt]	3.12		4.08		([x0.667] x0.667)		([x0.236] x0.236)	6.40		8.28		([x1.01] x1.01)		([x0.578] x0.578)	15.7		49.5		([x1.18] x1.18)		([x0.736] x0.736)						
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	4.70		4.55		x1.04		x0.764	5.23		5.08		x2.62		x1.92	6.70		9.47		x10.4		x6.61						
		run [μ s/pt]	3.53		3.98		([x0.401] x0.401)		([x0.221] x0.221)	5.83		8.99		([x0.764] x0.764)		([x0.537] x0.537)	11.0		51.1		([x1.31] x1.31)		([x0.747] x0.746)						
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	4.22		4.15		x1.26		x0.932	4.73		4.38		x1.95		x1.48	5.59		5.37		x4.96		x4.15						
		run [μ s/pt]	1.17		1.03		([x0.640] x0.640)		([x0.395] x0.395)	2.91		4.59		([x0.643] x0.643)		([x0.217] x0.217)	5.87		9.32		([x0.973] x0.973)		([x0.574] x0.573)						
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	4.34		4.39		x0.705		x0.540	4.45		4.45		x1.09		x0.807	8.28		5.31		x1.52		x1.83						
		run [μ s/pt]	0.623		0.921		([x0.653] x0.653)		([x0.400] x0.400)	3.04		3.97		([x0.349] x0.349)		([x0.250] x0.250)	6.05		8.50		([x0.578] x0.578)		([x0.591] x0.591)						
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	4.80		4.29		x1.02		x0.723	6.59		4.60		x1.70		x1.48	9.02		6.51		x4.87		x4.59						
		run [μ s/pt]	4.00		1.93		([x0.743] x0.743)		([x0.280] x0.280)	9.90		8.56		([x0.798] x0.798)		([x0.189] x0.189)	24.0		18.0		([x0.970] x0.970)		([x0.462] x0.462)						
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	11.1		4.70		x1.67		x1.55	36.2		7.66		x2.67		x5.99	125		28.7		x1.63		x15.6						
		run [μ s/pt]	5.08		5.67		([x0.776] x0.775)		([x0.703] x0.703)	12.9		27.3		([x0.985] x0.985)		([x1.22] x1.22)	33.4		228		([x1.12] x1.12)		([x1.97] x1.97)						
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	6.20		4.53		x1.80		x1.23	9.88		5.29		x3.44		x3.74	23.9		11.6		x1.60		x10.2						
		run [μ s/pt]	3.02		5.23		([x0.464] x0.464)		([x0.235] x0.235)	6.02		11.1		([x0.773] x0.773)		([x0.604] x0.604)	15.3		74.8		([x0.924] x0.924)		([x0.690] x0.689)						
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	9.48		6.11		x2.43		x2.14	24.7		20.4		x1.91		x5.25	76.0		192		x1.29		>15GB						
		run [μ s/pt]	4.11		11.6		([x1.19] x1.19)		([x0.800] x0.799)	11.1		94.5		([x1.63] x1.63)		([x1.26] x1.26)	29.9		1.60e+03		([x1.69] x1.69)		>15GB						
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.58		4.38		x0.992		x0.654	19.3		4.78		x0.876		x1.72	12.7		5.62		x2.65		x2.64						
		run [μ s/pt]	8.14		1.94		([x0.866] x0.866)		([x0.297] x0.296)	45.7		11.3		([x0.712] x0.711)		([x0.114] x0.114)	44.4		9.27		([x0.970] x0.969)		([x0.393] x0.392)						
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	4.49		4.25		x0.903		x0.769	4.68		4.46		x1.07		x0.857	5.21		4.54		x1.82		x1.23						
		run [μ s/pt]	2.49		1.26		([x0.527] x0.527)		([x0.379] x0.379)	3.44		1.96		([x0.770] x0.770)		([x0.379] x0.379)	6.59		4.33		([x0.823] x0.823)		([x0.282] x0.282)						
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	5.25		4.66		x0.891		x0.643	7.58		4.45		x1.66		x1.10	16.9		5.30		x3.01		x2.63						
		run [μ s/pt]	3.66		1.80		([x0.554] x0.554)		([x0.383] x0.383)	8.97		5.09		([x0.664] x0.663)		([x0.361] x0.361)	23.5		11.2		([x0.745] x0.744)		([x0.605] x0.605)						
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	4.87		4.65		x2.37		x2.28	5.41		4.81		x1.84		x1.77	5.51		4.48		x3.31		x2.62						
		run [μ s/pt]	2.85		2.86		([x1.09] x1.09)		([x0.240] x0.239)	3.95		3.44		([x0.798] x0.797)		([x0.251] x0.251)	7.10		6.17		([x0.777] x0.776)		([x0.225] x0.224)						
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	5.27		4.58		x1.37		x1.25	6.84		5.13		x3.22		x2.83	26.3		8.75		x4.75		x6.73						
		run [μ s/pt]	0.771		2.97		([x0.733] x0.733)		([x0.250] x0.250)	1.73		6.10		([x0.606] x0.606)		([x0.479] x0.479)	4.16		31.8		([x0.753] x0.753)		([x0.442] x0.442)						
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE						NOT APPLICABLE																				
		run [μ s/pt]													9.13 5.83 x6.89 x7.88														
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	4.99		5.69		x2.71		x1.50	7.37		6.02		x5.92		x3.91	22.7		9.33		x7.28		x10.5						
		run [μ s/pt]	1.52		3.02		([x0.463] x0.461)		([x0.406] x0.406)	2.16		10.4		([x0.543] x0.543)		([x0.479] x0.478)	4.71		60.0		([x0.688] x0.687)		([x0.400] x0.399)						
summary: generation			x0.705		x2.71		x1.28		x1.46		x1.54	x0.876		x5.92		x1.93		x2.39		x2.33	x1.29		x14.1		x3.31		x4.67		x2.96
summary: run single-flow, hel. sum			x0.401		x1.19		x0.660		x0.697		x0.740	x0.349		x1.63		x0.767		x0.789		x0.836	x0.578		x1.69		x0.969		x0.971		x1.06
summary: run all-flows, single hel.			x0.221		x0.799		x0.338		x0.373		x0.456	x0.114		x1.26		x0.429		x0.494		x0.900	x0.224		x1.97		x0.520		x0.613		x1.13

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. Runtime comparisons use (xS)xW. The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM compiled JIT O3 versus recurrence JIT O2 LC (block 3 of 3)

ID	base process	metric		n=7			n=8			n=9						
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	15.1		59.2		x1.54		x19.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	39.3		489		([x1.37] x1.37)		([x0.915] x0.915)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	11.7		58.5		x1.52		x11.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	27.4		485		([x1.63] x1.63)		([x1.04] x1.04)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	7.23		10.2		x20.2		x13.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	14.0		53.7		([x1.16] x1.16)		([x0.719] x0.719)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	6.37		10.1		x10.4		x5.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	8.96		50.5		([x1.22] x1.22)		([x0.735] x0.735)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	15.6		19.8		x17.6		x12.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	55.6		111		([x1.38] x1.38)		([x0.517] x0.517)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	463		373		x1.51		>15GB	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	89.0		4.55e+03		([x1.17] x1.17)		>15GB	N/A	N/A	N/A	N/A	N/A	N/A	N/A
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	67.3		91.8		x1.56		x11.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	39.8		1.01e+03		([x1.20] x1.19)		([x0.555] x0.555)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]	417		2.88e+03		x0.800		>15GB	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	76.1		4.25e+04		([x1.92] x1.90)		>15GB	N/A	N/A	N/A	N/A	N/A	N/A	N/A
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	21.9		12.1		x6.36		x7.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	108		36.5		([x1.05] x1.05)		([x0.410] x0.409)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	6.44		4.93		x3.19		x2.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	13.9		7.70		([x0.879] x0.879)		([x0.440] x0.440)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	45.4		9.14		x4.08		x7.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	60.6		55.3		([x0.774] x0.773)		([x0.653] x0.652)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	6.48		5.18		x5.96		x5.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	14.0		9.15		([x0.881] x0.881)		([x0.406] x0.406)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	97.1		32.9		x1.98		x9.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	10.9		285		([x0.679] x0.678)		([x0.335] x0.335)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	55.9		15.2		x6.36		x13.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	7.17		113		([x1.23] x1.23)		([x0.338] x0.338)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	110		82.5		x2.11		x7.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		run [μ s/pt]	17.0		1.05e+03		([x0.577] x0.576)		([x0.196] x0.195)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
summary: generation		x0.800			x20.2		x3.19		x5.67		x2.10		N/A		N/A	
summary: run single-flow, hel. sum		x0.576			x1.90		x1.17		x1.14		x1.22		N/A		N/A	
summary: run all-flows, single hel.		x0.195			x1.04		x0.517		x0.558		x0.544		N/A		N/A	

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. Runtime comparisons use (xS)xW. The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.11 Built-in SM compiled JIT O3 versus recurrence JIT O2 NLC

Built-in SM compiled JIT O3 versus recurrence JIT O2 NLC (block 1 of 2)

ID	base process	metric	n=1					n=2					n=3				
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	4.34	x0.481				4.49	x0.538				4.59	x0.647			
		run [μ s/pt]	0.166	([x0.872] x0.872)				0.489	([x0.667] x0.667)				1.68	([x0.718] x0.717)			
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	4.66	x0.438				4.20	x0.531				4.77	x0.513			
		run [μ s/pt]	0.114	([x1.17] x1.17)				0.296	([x0.906] x0.906)				1.26	([x0.658] x0.658)			
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]		NOT APPLICABLE				4.24	x0.585				4.23	x0.672			
		run [μ s/pt]						0.271	([x0.752] x0.752)				0.443	([x0.841] x0.841)			
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]		NOT APPLICABLE				4.45	x0.479				4.58	x0.506			
		run [μ s/pt]						0.117	([x1.35] x1.35)				0.198	([x1.31] x1.31)			
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]		NOT APPLICABLE				4.24	x0.576				4.36	x0.590			
		run [μ s/pt]						0.601	([x0.653] x0.653)				2.05	([x0.503] x0.503)			
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				4.43	x0.510				5.41	x0.700			
		run [μ s/pt]						1.77	([x0.338] x0.338)				8.06	([x0.700] x0.700)			
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				4.30	x0.565				4.85	x0.757			
		run [μ s/pt]						0.479	([x1.56] x1.56)				2.24	([x1.77] x1.77)			
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]		NOT APPLICABLE				5.14	x0.475				5.79	x0.623			
		run [μ s/pt]						10.0	([x0.152] x0.152)				33.8	([x0.893] x0.893)			
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE				4.72	x0.687			
		run [μ s/pt]											3.53	([x0.695] x0.695)			
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
summary: generation			x0.438	x0.481	x0.460	x0.460	<u>x0.459</u>	x0.475	x0.585	x0.535	x0.532	<u>x0.531</u>	x0.506	x0.757	x0.647	x0.633	<u>x0.634</u>
summary: wall			x0.872	x1.17	x1.02	x1.02	<u>x0.992</u>	x0.152	x1.56	x0.709	x0.797	<u>x0.300</u>	x0.503	x1.77	x0.717	x0.898	<u>x0.862</u>

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM compiled JIT O3 versus recurrence JIT O2 NLC (block 2 of 2)

ID	base process	metric		n=4			n=5			n=6		
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	6.79	x0.878		27.3	x1.50		NOT APPLICABLE			
		run [μ s/pt]	12.1	([x0.916]	x0.915)	126	([x1.23]	x1.23)				
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	5.63	x0.796		17.9	x1.52		NOT APPLICABLE			
		run [μ s/pt]	6.85	([x1.04]	x1.04)	67.5	([x1.37]	x1.37)				
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	4.50	x0.790		6.22	x1.09		NOT APPLICABLE			
		run [μ s/pt]	1.85	([x0.613]	x0.613)	10.3	([x0.936]	x0.936)				
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	4.29	x0.624		5.07	x0.778		NOT APPLICABLE			
		run [μ s/pt]	0.811	([x0.782]	x0.782)	5.65	([x0.871]	x0.870)				
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	5.69	x0.751		13.7	x1.17		NOT APPLICABLE			
		run [μ s/pt]	7.16	([x0.790]	x0.790)	53.6	([x0.974]	x0.973)				
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	15.9	x1.43		146	x2.17		NOT APPLICABLE			
		run [μ s/pt]	86.6	([x0.852]	x0.852)	3.23e+03	([x0.453]	x0.453)				
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.12	x1.87		31.2	x4.78		NOT APPLICABLE			
		run [μ s/pt]	16.5	([x2.15]	x2.15)	252	([x2.20]	x2.20)				
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	10.7	x0.879		28.6	x1.41		NOT APPLICABLE			
		run [μ s/pt]	517	([x1.29]	x1.29)	1.01e+04	([x1.43]	x1.43)				
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.87	x0.760		10.8	x2.68		NOT APPLICABLE			
		run [μ s/pt]	8.28	([x0.806]	x0.806)	51.6	([x1.60]	x1.60)				
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	4.66	x0.794		4.63	x0.977		NOT APPLICABLE			
		run [μ s/pt]	2.44	([x0.513]	x0.512)	3.47	([x0.732]	x0.732)				
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	5.31	x0.945		7.12	x2.33		NOT APPLICABLE			
		run [μ s/pt]	5.71	([x1.43]	x1.43)	29.2	([x1.75]	x1.75)				
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	4.52	x1.38		5.46	x1.57		NOT APPLICABLE			
		run [μ s/pt]	2.68	([x0.509]	x0.509)	3.72	([x0.737]	x0.736)				
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	5.14	x1.03		7.32	x2.26		41.3	x3.25		
		run [μ s/pt]	2.07	([x1.41]	x1.41)	15.5	([x1.35]	x1.35)	201	([x1.38]	x1.38)	
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE			NOT APPLICABLE			17.6	x4.83		
		run [μ s/pt]							43.7	([x1.44]	x1.44)	
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	5.07	x1.49		10.3	x2.66		71.1	x3.06		
		run [μ s/pt]	3.30	([x1.38]	x1.38)	27.4	([x1.24]	x1.24)	504	([x0.935]	x0.935)	
summary: generation		x0.624 x1.87 x0.878 x1.03 [x1.08]			x0.778 x4.78 x1.55 x1.92 [x2.19]			x3.06 x4.83 x3.25 x3.71 [x3.36]				
summary: wall		x0.509 x2.15 x0.884 x1.03 [x1.23]			x0.453 x2.20 x1.23 x1.20 [x1.21]			x0.935 x1.44 x1.38 x1.25 [x1.08]				

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.12 Built-in SM compiled JIT O3 versus recurrence JIT O2 full-colour

Built-in SM compiled JIT O3 versus recurrence JIT O2 full-colour (block 1 of 2)

ID	base process	metric	n=1					n=2					n=3				
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	4.47	x0.523				4.37	x0.567				5.08	x0.552			
		run [μ s/pt]	0.171	([x0.845] x0.845)				0.459	([x0.717] x0.716)				1.73	([x0.732] x0.732)			
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	4.32	x0.467				4.36	x0.538				4.70	x0.512			
		run [μ s/pt]	0.109	([x1.20] x1.20)				0.283	([x1.01] x1.01)				1.42	([x0.572] x0.572)			
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]		NOT APPLICABLE				4.49	x0.591				4.52	x0.675			
		run [μ s/pt]						0.282	([x0.755] x0.755)				0.469	([x0.839] x0.839)			
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]		NOT APPLICABLE				4.50	x0.486				4.49	x0.518			
		run [μ s/pt]						0.117	([x1.35] x1.35)				0.206	([x1.25] x1.25)			
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]		NOT APPLICABLE				4.24	x0.529				4.92	x0.612			
		run [μ s/pt]						0.595	([x0.654] x0.654)				2.13	([x0.494] x0.494)			
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				4.65	x0.502				5.64	x0.689			
		run [μ s/pt]						2.02	([x0.303] x0.303)				8.52	([x0.737] x0.736)			
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				4.31	x0.569				4.75	x0.783			
		run [μ s/pt]						0.505	([x1.48] x1.48)				2.29	([x1.70] x1.70)			
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]		NOT APPLICABLE				8.38	x0.301				5.87	x0.669			
		run [μ s/pt]						12.2	([x0.149] x0.149)				34.5	([x0.887] x0.887)			
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE				4.84	x0.662			
		run [μ s/pt]											3.73	([x0.661] x0.661)			
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
summary: generation			n=1					n=2					n=3				
summary: wall			x0.467	x0.523	x0.495	x0.495	x0.495	x0.301	x0.591	x0.534	x0.510	x0.489	x0.512	x0.783	x0.662	x0.630	x0.632
			x0.845	x1.20	x1.02	x1.02	x0.983	x0.149	x1.48	x0.736	x0.803	x0.277	x0.494	x1.70	x0.736	x0.875	x0.855

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM compiled JIT O3 versus recurrence JIT O2 full-colour (block 2 of 2)

ID	base process	metric		n=4			n=5			n=6								
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	6.80	x0.885		28.1	x1.39		NOT APPLICABLE									
		run [μ s/pt]	12.4	([x0.903]	x0.902)	133	([x1.21]	x1.21)										
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	5.86	x0.802		17.7	x1.55		NOT APPLICABLE									
		run [μ s/pt]	7.26	([x0.984]	x0.984)	68.0	([x1.56]	x1.56)										
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	4.75	x0.785		6.26	x1.09		NOT APPLICABLE									
		run [μ s/pt]	1.93	([x0.600]	x0.600)	10.7	([x0.897]	x0.896)										
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	4.53	x0.567		4.96	x0.801		NOT APPLICABLE									
		run [μ s/pt]	0.804	([x0.774]	x0.774)	5.79	([x0.842]	x0.841)										
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	5.65	x0.739		14.4	x1.08		NOT APPLICABLE									
		run [μ s/pt]	7.38	([x0.757]	x0.757)	56.1	([x0.913]	x0.913)										
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	15.9	x1.42		146	x2.09		NOT APPLICABLE									
		run [μ s/pt]	86.6	([x0.930]	x0.930)	3.30e+03	([x0.615]	x0.615)										
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.24	x1.83		32.8	x3.77		NOT APPLICABLE									
		run [μ s/pt]	16.5	([x2.18]	x2.18)	263	([x2.16]	x2.16)										
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	10.8	x0.919		32.9	x2.73		NOT APPLICABLE									
		run [μ s/pt]	549	([x1.23]	x1.23)	1.21e+04	([x1.32]	x1.32)										
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.69	x0.788		10.5	x1.22		NOT APPLICABLE									
		run [μ s/pt]	8.71	([x0.759]	x0.759)	36.6	([x2.98]	x2.98)										
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	4.58	x0.933		4.82	x0.918		NOT APPLICABLE									
		run [μ s/pt]	2.35	([x0.531]	x0.531)	3.39	([x0.750]	x0.750)										
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	4.88	x1.07		7.02	x2.29		NOT APPLICABLE									
		run [μ s/pt]	5.66	([x1.41]	x1.41)	28.8	([x1.63]	x1.63)										
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	4.52	x1.41		4.71	x1.71		NOT APPLICABLE									
		run [μ s/pt]	2.53	([x0.542]	x0.542)	3.59	([x0.742]	x0.741)										
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	4.91	x1.14		7.83	x2.21		52.5	x2.56								
		run [μ s/pt]	2.06	([x1.42]	x1.42)	16.0	([x1.34]	x1.34)	327	([x0.944]	x0.943)							
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE			NOT APPLICABLE			17.7	x4.81								
		run [μ s/pt]							45.3	([x1.39]	x1.39)							
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	4.94	x1.50		10.2	x2.86		71.3	x3.15								
		run [μ s/pt]	3.22	([x1.37]	x1.37)	27.0	([x1.25]	x1.25)	511	([x0.960]	x0.960)							
summary: generation				n=4			n=5			n=6								
summary: wall				x0.567	x1.83	x0.926	x1.06	x1.10	x0.801	x3.77	x1.63	x1.84	x2.13	x2.56	x4.81	x3.15	x3.51	x3.14
				x0.531	x2.18	x0.916	x1.03	x1.19	x0.615	x2.98	x1.23	x1.30	x1.19	x0.943	x1.39	x0.960	x1.10	x0.975

Baseline: recurrence JIT O2; candidate: compiled JIT O3. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.13 Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 LC

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 LC (block 1 of 3)

ID	base process	metric	n=1					n=2					n=3							
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	4.50		4.24	x0.464		x0.499	4.46		4.25	x0.536		x0.493	4.92		5.08	x0.466		x0.444
		run [μ s/pt]	0.169		0.0924	([x1.67] x1.67)		([x1.87] x1.87)	0.470		0.258	([x1.56] x1.56)		([x1.20] x1.20)	1.30		0.843	([x1.18] x1.18)		([x0.723] x0.723)
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	4.33		4.35	x0.553		x0.523	4.43		4.40	x0.477		x0.494	4.38		4.47	x0.540		x0.501
		run [μ s/pt]	0.117		0.0934	([x1.99] x1.99)		([x1.94] x1.94)	0.300		0.260	([x1.54] x1.53)		([x1.12] x1.12)	0.835		0.845	([x1.19] x1.18)		([x0.679] x0.678)
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE					4.49		4.45	x0.496		x0.525	4.45		4.41	x0.502		x0.515	
		run [μ s/pt]						0.295		0.243	([x1.24] x1.24)		([x1.02] x1.02)	0.512		0.422	([x1.41] x1.41)		([x0.845] x0.844)	
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE					4.42		4.45	x0.503		x0.487	4.50		4.48	x0.478		x0.484	
		run [μ s/pt]						0.135		0.189	([x1.55] x1.55)		([x1.09] x1.09)	0.232		0.343	([x1.50] x1.50)		([x0.898] x0.897)	
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE					4.22		4.04	x0.483		x0.565	4.38		4.23	x0.495		x0.512	
		run [μ s/pt]						0.593		0.282	([x1.58] x1.57)		([x1.16] x1.16)	2.10		0.660	([x1.30] x1.30)		([x0.794] x0.794)	
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE					4.38		4.15	x0.526		x0.518	5.39		4.43	x0.458		x0.572	
		run [μ s/pt]						1.04		0.400	([x0.687] x0.686)		([x1.12] x1.12)	2.37		2.13	([x1.08] x1.08)		([x0.761] x0.761)	
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE					4.28		4.21	x0.499		x0.508	5.17		4.83	x0.457		x0.485	
		run [μ s/pt]						0.290		0.221	([x1.53] x1.53)		([x1.39] x1.39)	1.28		1.16	([x1.06] x1.06)		([x0.663] x0.662)	
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	NOT APPLICABLE					4.78		4.55	x0.467		x0.517	5.53		4.50	x0.466		x0.615	
		run [μ s/pt]						1.52		1.53	([x0.435] x0.435)		([x0.383] x0.383)	2.99		3.65	([x0.991] x0.991)		([x0.773] x0.773)	
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					4.91		4.26	x0.486		x0.565		
		run [μ s/pt]											3.73		0.868	([x1.42] x1.42)		([x0.810] x0.810)		
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE							
		run [μ s/pt]																		
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE							
		run [μ s/pt]																		
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE							
		run [μ s/pt]																		
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE							
		run [μ s/pt]																		
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE							
		run [μ s/pt]																		
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE							
		run [μ s/pt]																		
summary: generation			x0.464		x0.553		x0.509		x0.509		x0.508	x0.467		x0.536		x0.497		x0.498		x0.498
summary: run single-flow, hel. sum			x1.67		x1.99		x1.83		x1.83		x1.80	x0.435		x1.57		x1.53		x1.26		x0.972
summary: run all-flows, single hel.			x1.87		x1.94		x1.91		x1.91		x1.91	x0.383		x1.39		x1.12		x1.06		x0.806

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 LC (block 2 of 3)

ID	base process	metric	n=4					n=5					n=6													
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	4.82		4.33	x0.488		x0.772	5.50		4.95	x0.554		x3.63	7.76		9.39	x0.536		x31.4						
		run [μ s/pt]	3.12		4.08	([x1.42] x1.42)		([x0.521] x0.520)	6.40		8.28	([x2.01] x2.01)		([x1.52] x1.52)	15.7		49.5	([x2.18] x2.18)		([x2.20] x2.20)						
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	4.70		4.55	x0.492		x0.655	5.23		5.08	x0.506		x2.17	6.70		9.47	x0.542		x17.7						
		run [μ s/pt]	3.53		3.98	([x0.844] x0.844)		([x0.484] x0.484)	5.83		8.99	([x1.51] x1.51)		([x1.12] x1.12)	11.0		51.1	([x2.24] x2.24)		([x1.51] x1.51)						
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	4.22		4.15	x0.517		x0.515	4.73		4.38	x0.505		x0.725	5.59		5.37	x0.498		x2.50						
		run [μ s/pt]	1.17		1.03	([x1.26] x1.26)		([x0.622] x0.622)	2.91		4.59	([x1.40] x1.40)		([x0.459] x0.459)	5.87		9.32	([x1.97] x1.97)		([x1.21] x1.21)						
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	4.34		4.39	x0.485		x0.529	4.45		4.45	x0.513		x0.607	8.28		5.31	x0.318		x1.62						
		run [μ s/pt]	0.623		0.921	([x1.17] x1.17)		([x0.678] x0.678)	3.04		3.97	([x0.722] x0.722)		([x0.441] x0.440)	6.05		8.50	([x1.12] x1.12)		([x1.02] x1.02)						
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	4.80		4.29	x0.531		x0.660	6.59		4.60	x0.484		x1.62	9.02		6.51	x0.558		x10.9						
		run [μ s/pt]	4.00		1.93	([x1.55] x1.55)		([x0.634] x0.633)	9.90		8.56	([x1.65] x1.65)		([x0.502] x0.502)	24.0		18.0	([x1.88] x1.87)		([x1.55] x1.55)						
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	11.1		4.70	x0.421		x1.56	36.2		7.66	x0.423		x12.6	125		28.7	x0.461		>15GB						
		run [μ s/pt]	5.08		5.67	([x1.66] x1.66)		([x1.68] x1.68)	12.9		27.3	([x2.00] x1.99)		([x3.04] x3.04)	33.4		228	([x2.20] x2.19)		>15GB						
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	6.20		4.53	x0.425		x0.678	9.88		5.29	x0.427		x2.23	23.9		11.6	x0.334		x15.0						
		run [μ s/pt]	3.02		5.23	([x1.09] x1.09)		([x0.560] x0.559)	6.02		11.1	([x1.67] x1.67)		([x1.49] x1.48)	15.3		74.8	([x1.87] x1.87)		([x1.88] x1.88)						
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	9.48		6.11	x0.456		x2.05	24.7		20.4	x0.473		x10.5	76.0		192	x0.374		>15GB						
		run [μ s/pt]	4.11		11.6	([x2.53] x2.53)		([x2.05] x2.05)	11.1		94.5	([x2.86] x2.86)		([x2.16] x2.16)	29.9		1.60e+03	([x2.83] x2.83)		>15GB						
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.58		4.38	x0.519		x0.655	19.3		4.78	x0.309		x1.28	12.7		5.62	x0.720		x5.20						
		run [μ s/pt]	8.14		1.94	([x1.74] x1.74)		([x0.577] x0.577)	45.7		11.3	([x0.754] x0.754)		([x0.284] x0.284)	44.4		9.27	([x1.79] x1.79)		([x1.08] x1.08)						
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	4.49		4.25	x0.495		x0.525	4.68		4.46	x0.508		x0.536	5.21		4.54	x0.516		x0.673						
		run [μ s/pt]	2.49		1.26	([x0.998] x0.998)		([x0.602] x0.602)	3.44		1.96	([x1.45] x1.44)		([x0.529] x0.529)	6.59		4.33	([x1.56] x1.56)		([x0.451] x0.451)						
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	5.25		4.66	x0.464		x0.523	7.58		4.45	x0.460		x0.788	16.9		5.30	x0.443		x2.35						
		run [μ s/pt]	3.66		1.80	([x1.02] x1.02)		([x0.658] x0.658)	8.97		5.09	([x1.22] x1.22)		([x0.631] x0.630)	23.5		11.2	([x1.22] x1.22)		([x1.11] x1.11)						
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	4.87		4.65	x0.542		x0.618	5.41		4.81	x0.549		x0.549	5.51		4.48	x0.512		x0.705						
		run [μ s/pt]	2.85		2.86	([x1.00] x1.00)		([x0.329] x0.329)	3.95		3.44	([x1.42] x1.42)		([x0.364] x0.364)	7.10		6.17	([x1.50] x1.50)		([x0.367] x0.366)						
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	5.27		4.58	x0.462		x0.562	6.84		5.13	x0.493		x0.651	26.3		8.75	x0.352		x1.74						
		run [μ s/pt]	0.771		2.97	([x1.40] x1.40)		([x0.412] x0.412)	1.73		6.10	([x1.79] x1.79)		([x0.771] x0.771)	4.16		31.8	([x1.65] x1.65)		([x0.758] x0.757)						
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					9.13					x0.571		x0.889						
		run [μ s/pt]											3.64					([x1.98] x1.98)		([x0.610] x0.609)						
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	4.99		5.69	x0.520		x0.517	7.37		6.02	x0.462		x0.690	22.7		9.33	x0.397		x2.35						
		run [μ s/pt]	1.52		3.02	([x0.755] x0.752)		([x0.606] x0.606)	2.16		10.4	([x1.22] x1.22)		([x0.755] x0.754)	4.71		60.0	([x1.52] x1.52)		([x0.676] x0.676)						
summary: generation			x0.421		x0.542	x0.490		x0.487		x0.479	x0.309		x0.554	x0.489		x0.476		x0.447	x0.318		x0.720	x0.498		x0.475		x0.438
summary: run single-flow, hel. sum			x0.752		x2.53	x1.21		x1.32		x1.44	x0.722		x2.86	x1.48		x1.55		x1.40	x1.12		x2.83	x1.87		x1.83		x1.95
summary: run all-flows, single hel.			x0.329		x2.05	x0.604		x0.743		x1.03	x0.284		x3.04	x0.692		x1.00		x1.73	x0.366		x2.20	x1.08		x1.11		x1.36

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 LC (block 3 of 3)

ID	base process	metric	n=7					n=8					n=9								
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	15.1		59.2		x0.561		>15GB	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	39.3		489	⌈x2.37⌋	x2.37		>15GB	N/A		N/A	N/A		N/A	N/A		N/A			
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	11.7		58.5		x0.497		>15GB	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	27.4		485	⌈x2.39⌋	x2.39		>15GB	N/A		N/A	N/A		N/A	N/A		N/A			
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	7.23		10.2		x0.530		x19.2	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	14.0		53.7	⌈x2.17⌋	x2.17		⌈x1.59⌋	x1.59		N/A	N/A		N/A	N/A		N/A			
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	6.37		10.1		x0.558		x11.2	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	8.96		50.5	⌈x2.26⌋	x2.26		⌈x1.28⌋	x1.27		N/A	N/A		N/A	N/A		N/A			
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	15.6		19.8		x0.522		>15GB	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	55.6		111	⌈x1.86⌋	x1.86		>15GB	N/A		N/A	N/A		N/A	N/A		N/A			
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	463		373		x0.470		>15GB	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	89.0		4.55e+03	⌈x2.14⌋	x2.13		>15GB	N/A		N/A	N/A		N/A	N/A		N/A			
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	67.3		91.8		x0.287		>15GB	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	39.8		1.01e+03	⌈x1.90⌋	x1.89		>15GB	N/A		N/A	N/A		N/A	N/A		N/A			
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	417		2.88e+03		x0.228		>15GB	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	76.1		4.25e+04	⌈x2.82⌋	x2.79		>15GB	N/A		N/A	N/A		N/A	N/A		N/A			
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	21.9		12.1		x0.634		x32.6	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	108		36.5	⌈x1.68⌋	x1.68		⌈x1.80⌋	x1.80		N/A	N/A		N/A	N/A		N/A			
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	6.44		4.93		x0.544		x1.74	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	13.9		7.70	⌈x1.70⌋	x1.69		⌈x0.753⌋	x0.753		N/A	N/A		N/A	N/A		N/A			
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	45.4		9.14		x0.454		x13.8	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	60.6		55.3	⌈x1.27⌋	x1.27		⌈x1.22⌋	x1.22		N/A	N/A		N/A	N/A		N/A			
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	6.48		5.18		x0.546		x1.70	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	14.0		9.15	⌈x1.71⌋	x1.71		⌈x0.717⌋	x0.716		N/A	N/A		N/A	N/A		N/A			
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	97.1		32.9		x0.282		x5.79	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	10.9		285	⌈x1.76⌋	x1.75		⌈x0.669⌋	x0.669		N/A	N/A		N/A	N/A		N/A			
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	55.9		15.2		x0.614		x2.22	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	7.17		113	⌈x4.05⌋	x4.04		⌈x0.492⌋	x0.492		N/A	N/A		N/A	N/A		N/A			
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	110		82.5		x0.265		x5.33	N/A		N/A	N/A		N/A	N/A		N/A			
		run [μ s/pt]	17.0		1.05e+03	⌈x1.18⌋	x1.18		⌈x0.311⌋	x0.310		N/A	N/A		N/A	N/A		N/A			
summary: generation			x0.228		x0.634		x0.522		x0.466		⌈x0.367⌋	n=8					n=9				
summary: run single-flow, hel. sum			x1.18		x4.04		x1.89		x2.08		⌈x2.00⌋	N/A					N/A				
summary: run all-flows, single hel.			x0.310		x1.80		x0.753		x0.980		⌈x0.522⌋	N/A					N/A				

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the topology-replay/all-flow-union baseline generation times and wall times, followed by layout-matched candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums and is the framed bold entry. LC generation uses selected flow only, while LC runtime shows separate selected-flow and all-flow wall-only lines. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.14 Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 NLC

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 NLC (block 1 of 2)

ID	base process	metric	n=1					n=2					n=3				
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	4.34	x0.477				4.49	x0.468				4.59	x0.500			
		run [μ s/pt]	0.166	([x1.13] x1.13)				0.489	([x1.07] x1.07)				1.68	([x1.35] x1.35)			
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	4.66	x0.450				4.20	x0.498				4.77	x0.471			
		run [μ s/pt]	0.114	([x1.42] x1.42)				0.296	([x1.38] x1.38)				1.26	([x1.23] x1.23)			
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]		NOT APPLICABLE				4.24	x0.524				4.23	x0.557			
		run [μ s/pt]						0.271	([x1.09] x1.09)				0.443	([x1.41] x1.41)			
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]		NOT APPLICABLE				4.45	x0.494				4.58	x0.480			
		run [μ s/pt]						0.117	([x1.52] x1.52)				0.198	([x1.58] x1.58)			
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]		NOT APPLICABLE				4.24	x0.537				4.36	x0.523			
		run [μ s/pt]						0.601	([x1.10] x1.10)				2.05	([x1.02] x1.02)			
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				4.43	x0.515				5.41	x0.510			
		run [μ s/pt]						1.77	([x0.556] x0.556)				8.06	([x0.994] x0.994)			
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]		NOT APPLICABLE				4.30	x0.494				4.85	x0.483			
		run [μ s/pt]						0.479	([x0.885] x0.885)				2.24	([x1.14] x1.14)			
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]		NOT APPLICABLE				5.14	x0.436				5.79	x0.455			
		run [μ s/pt]						10.0	([x0.311] x0.310)				33.8	([x1.76] x1.75)			
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE				4.72	x0.501			
		run [μ s/pt]											3.53	([x1.24] x1.24)			
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]		NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE			
		run [μ s/pt]															
summary: generation			n=1					n=2					n=3				
summary: wall			x0.450	x0.477	x0.464	x0.464	x0.463	x0.436	x0.537	x0.496	x0.496	x0.494	x0.455	x0.557	x0.500	x0.498	x0.496
			x1.13	x1.42	x1.27	x1.27	x1.25	x0.310	x1.52	x1.08	x0.990	x0.469	x0.994	x1.75	x1.24	x1.30	x1.52

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 NLC (block 2 of 2)

ID	base process	metric		n=4			n=5			n=6									
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	6.79	x0.549			27.3	x0.591			NOT APPLICABLE								
		run [μ s/pt]	12.1	([x1.44]	x1.44)		126	([x1.48]	x1.48)										
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	5.63	x0.527			17.9	x0.633			NOT APPLICABLE								
		run [μ s/pt]	6.85	([x1.72]	x1.72)		67.5	([x1.90]	x1.90)										
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	4.50	x0.488			6.22	x0.504			NOT APPLICABLE								
		run [μ s/pt]	1.85	([x1.18]	x1.18)		10.3	([x1.59]	x1.59)										
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	4.29	x0.509			5.07	x0.535			NOT APPLICABLE								
		run [μ s/pt]	0.811	([x1.26]	x1.26)		5.65	([x1.36]	x1.36)										
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	5.69	x0.498			13.7	x0.569			NOT APPLICABLE								
		run [μ s/pt]	7.16	([x1.50]	x1.49)		53.6	([x1.48]	x1.47)										
6	$gg \rightarrow t\bar{t} + (n-2)g$	gen. [s]	15.9	x0.650			146	x1.03			NOT APPLICABLE								
		run [μ s/pt]	86.6	([x1.08]	x1.08)		3.23e+03	([x0.442]	x0.442)										
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.12	x0.530			31.2	x0.671			NOT APPLICABLE								
		run [μ s/pt]	16.5	([x1.45]	x1.45)		252	([x1.20]	x1.20)										
8	$gg \rightarrow gg + (n-2)g$	gen. [s]	10.7	x0.460			28.6	x0.691			NOT APPLICABLE								
		run [μ s/pt]	517	([x2.23]	x2.23)		1.01e+04	([x2.16]	x2.16)										
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.87	x0.483			10.8	x1.52			NOT APPLICABLE								
		run [μ s/pt]	8.28	([x1.45]	x1.45)		51.6	([x2.58]	x2.58)										
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	4.66	x0.495			4.63	x0.509			NOT APPLICABLE								
		run [μ s/pt]	2.44	([x0.943]	x0.943)		3.47	([x1.32]	x1.32)										
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	5.31	x0.512			7.12	x0.566			NOT APPLICABLE								
		run [μ s/pt]	5.71	([x1.08]	x1.08)		29.2	([x1.15]	x1.15)										
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	4.52	x0.501			5.46	x0.468			NOT APPLICABLE								
		run [μ s/pt]	2.68	([x0.969]	x0.968)		3.72	([x1.35]	x1.35)										
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	5.14	x0.536			7.32	x0.649			41.3	x0.806							
		run [μ s/pt]	2.07	([x1.45]	x1.45)		15.5	([x1.64]	x1.64)		201	([x1.52] x1.51)							
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE				NOT APPLICABLE				17.6	x0.753							
		run [μ s/pt]									43.7	([x1.64] x1.64)							
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	5.07	x0.525			10.3	x0.581			71.1	x0.755							
		run [μ s/pt]	3.30	([x1.38]	x1.38)		27.4	([x1.58]	x1.58)		504	([x1.13] x1.13)							
summary: generation					n=4						n=5						n=6		
		x0.460	x0.650	x0.511	x0.519	x0.532	x0.468	x1.52	x0.586	x0.679	x0.833	x0.753	x0.806	x0.755	x0.771	x0.771			
summary: wall					n=4						n=5						n=6		
		x0.943	x2.23	x1.41	x1.37	x2.00	x0.442	x2.58	x1.48	x1.52	x1.73	x1.13	x1.64	x1.51	x1.43	x1.26			

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

5.15 Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 full-colour

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 full-colour (block 1 of 2)

ID	base process	metric	n=1					n=2					n=3				
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	4.47	x0.470				4.37	x0.503				5.08	x0.484			
		run [μ s/pt]	0.171	([x1.09] x1.09)				0.459	([x1.15] x1.15)				1.73	([x1.36] x1.36)			
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	4.32	x0.484				4.36	x0.505				4.70	x0.470			
		run [μ s/pt]	0.109	([x1.46] x1.46)				0.283	([x1.39] x1.39)				1.42	([x1.06] x1.06)			
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	NOT APPLICABLE					4.49	x0.523				4.52	x0.515			
		run [μ s/pt]						0.282	([x1.05] x1.05)				0.469	([x1.31] x1.31)			
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	NOT APPLICABLE					4.50	x0.491				4.49	x0.500			
		run [μ s/pt]						0.117	([x1.57] x1.57)				0.206	([x1.51] x1.51)			
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	NOT APPLICABLE					4.24	x0.482				4.92	x0.458			
		run [μ s/pt]						0.595	([x1.10] x1.10)				2.13	([x0.988] x0.988)			
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE					4.65	x0.523				5.64	x1.12			
		run [μ s/pt]						2.02	([x0.489] x0.488)				8.52	([x1.17] x1.17)			
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	NOT APPLICABLE					4.31	x0.489				4.75	x0.497			
		run [μ s/pt]						0.505	([x0.837] x0.836)				2.29	([x1.12] x1.12)			
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]	NOT APPLICABLE					8.38	x0.288				5.87	x0.485			
		run [μ s/pt]						12.2	([x0.267] x0.267)				34.5	([x1.90] x1.90)			
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					4.84	x0.493			
		run [μ s/pt]											3.73	([x1.18] x1.18)			
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE				
		run [μ s/pt]															
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE				
		run [μ s/pt]															
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE				
		run [μ s/pt]															
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE				
		run [μ s/pt]															
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE				
		run [μ s/pt]															
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	NOT APPLICABLE					NOT APPLICABLE					NOT APPLICABLE				
		run [μ s/pt]															
summary: generation			n=1					n=2					n=3				
summary: wall			x0.470	x0.484	x0.477	x0.477	x0.477	x0.288	x0.523	x0.497	x0.475	x0.457	x0.458	x1.12	x0.493	x0.558	x0.567
			x1.09	x1.46	x1.28	x1.28	x1.24	x0.267	x1.57	x1.07	x0.981	x0.408	x0.988	x1.90	x1.18	x1.29	x1.63

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

Built-in SM eager-DAG JIT O2 versus recurrence JIT O2 full-colour (block 2 of 2)

ID	base process	metric	n=4			n=5			n=6					
1	$d\bar{d} \rightarrow Z + (n-1)g$	gen. [s]	6.80	x0.500		28.1	x0.571		NOT APPLICABLE					
		run [μ s/pt]	12.4	([x1.43]	x1.42)	133	([x1.49]	x1.49)						
2	$u\bar{d} \rightarrow W^+ + (n-1)g$	gen. [s]	5.86	x0.516		17.7	x0.639		NOT APPLICABLE					
		run [μ s/pt]	7.26	([x1.63]	x1.63)	68.0	([x2.02]	x2.02)						
3	$d\bar{d} \rightarrow e^+e^- + (n-2)g$	gen. [s]	4.75	x0.502		6.26	x0.502		NOT APPLICABLE					
		run [μ s/pt]	1.93	([x1.36]	x1.36)	10.7	([x1.47]	x1.47)						
4	$u\bar{d} \rightarrow e^+\nu_e + (n-2)g$	gen. [s]	4.53	x0.514		4.96	x0.536		NOT APPLICABLE					
		run [μ s/pt]	0.804	([x1.27]	x1.27)	5.79	([x1.33]	x1.33)						
5	$d\bar{d} \rightarrow ZZ + (n-2)g$	gen. [s]	5.65	x0.499		14.4	x0.540		NOT APPLICABLE					
		run [μ s/pt]	7.38	([x1.35]	x1.35)	56.1	([x1.35]	x1.35)						
6	$g\bar{g} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	15.9	x0.653		146	x1.05		NOT APPLICABLE					
		run [μ s/pt]	86.6	([x1.15]	x1.15)	3.30e+03	([x0.576]	x0.576)						
7	$d\bar{d} \rightarrow t\bar{t} + (n-2)g$	gen. [s]	7.24	x0.529		32.8	x0.625		NOT APPLICABLE					
		run [μ s/pt]	16.5	([x1.56]	x1.56)	263	([x1.23]	x1.23)						
8	$g\bar{g} \rightarrow g\bar{g} + (n-2)g$	gen. [s]	10.8	x0.470		32.9	x1.21		NOT APPLICABLE					
		run [μ s/pt]	549	([x2.93]	x2.93)	1.21e+04	([x6.29]	x6.29)						
9	$d\bar{d} \rightarrow ZZZ + (n-3)g$	gen. [s]	5.69	x0.506		10.5	x1.47		NOT APPLICABLE					
		run [μ s/pt]	8.71	([x1.40]	x1.40)	36.6	([x3.61]	x3.62)						
10	$d\bar{d} \rightarrow e^+e^-ZH + (n-4)g$	gen. [s]	4.58	x0.550		4.82	x0.487		NOT APPLICABLE					
		run [μ s/pt]	2.35	([x0.967]	x0.966)	3.39	([x1.37]	x1.37)						
11	$d\bar{d} \rightarrow t\bar{t}ZH + (n-4)g$	gen. [s]	4.88	x0.545		7.02	x0.590		NOT APPLICABLE					
		run [μ s/pt]	5.66	([x1.11]	x1.11)	28.8	([x1.17]	x1.17)						
12	$d\bar{d} \rightarrow e^+e^-e^+e^- + (n-4)g$	gen. [s]	4.52	x0.504		4.71	x0.508		NOT APPLICABLE					
		run [μ s/pt]	2.53	([x1.03]	x1.03)	3.59	([x1.37]	x1.36)						
13	$d\bar{d} \rightarrow u\bar{u}s\bar{s} + (n-4)g$	gen. [s]	4.91	x0.506		7.83	x0.621		52.5	x0.613				
		run [μ s/pt]	2.06	([x1.43]	x1.43)	16.0	([x1.65]	x1.65)	327	([x0.949]	x0.948)			
14	$d\bar{d} \rightarrow u\bar{u}s\bar{s}c\bar{c} + (n-6)g$	gen. [s]	NOT APPLICABLE			NOT APPLICABLE			17.7	x0.754				
		run [μ s/pt]							45.3	([x1.58]	x1.58)			
15	$d\bar{d} \rightarrow u\bar{u}u\bar{u} + (n-4)g$	gen. [s]	4.94	x0.526		10.2	x0.603		71.3	x0.750				
		run [μ s/pt]	3.22	([x1.40]	x1.40)	27.0	([x1.52]	x1.52)	511	([x1.04]	x1.03)			
summary: generation			n=4			n=5			n=6					
			x0.470	x0.653	x0.510	x0.523	x0.534	x0.487	x1.47	x0.597	x0.711	x0.884	x0.613	x0.754
summary: wall			n=4			n=5			n=6					
			x0.966	x2.93	x1.38	x1.43	x2.56	x0.576	x6.29	x1.42	x1.89	x4.92	x0.948	x1.58

Baseline: recurrence JIT O2; candidate: eager-DAG JIT O2. Each cell shows the baseline generation and wall time, followed by candidate/baseline generation and wall-time ratios. Runtime comparisons use ([xS]xW). The muted supplementary xS first compares compatible exposed execution attributions; otherwise it compares authenticated evaluator totals. Only when original AmpliCol has no evaluator-total clock may a candidate evaluator total use the legacy direct execution clock (or its wall clock when direct execution is absent) as denominator. An evaluator total is never divided by a recurrence core. This evaluator-total fallback is diagnostic only; the execution/execution ratio remains the compatible supplementary attribution. The coloured xW is always the candidate/baseline wall multiplier and remains the primary figure of merit. When no compatible supplementary clock exists, only (xW) is shown. No clock is fabricated from another. Every displayed number uses exactly three significant digits. Unverified compiled or eager diagnostics show absolute generation and runtime values with a yellow unverified marker, but never enter ratios or summaries. In their runtime token the muted bracketed value is evaluator-total and the following value is outer wall time. Ratios and summaries require exact stored linkage to the current denominator. Summary rows contain multipliers only in min, max, median, average, and weighted-average order; the weighted average is the ratio of timing sums, is the framed bold entry, and runtime statistics use wall time only. Not applicable marks a process/multiplicity combination outside the process-family definition. Fixed-engine tables intentionally omit mode letters because the selected engine is named in the table heading.

6 Z-plus-Jets Evaluator Ladders

The dedicated $d\bar{d} \rightarrow Z + (n-1)g$ ladders compare the independent reference with compiled JIT O1, compiled JIT O3, ASM O3, C++ O3, eager-DAG JIT O2, and recurrence JIT O2 evaluator configurations. They retain final-state multiplicities $n = 1, \dots, 9$ for both the built-in model and the UFO-SM result set. Each row states its evaluator backend and optimization level explicitly; all other workload settings follow the benchmark contract above.

The eager and recurrence rows start from the corresponding portable JIT O2 prepared model. Each layout retains complete flow and helicity coverage; the selected-flow workload sums helicities, while the all-flow workload selects one helicity at runtime. Prepared-model creation is excluded from process-generation time. Every displayed Z -ladder timing ratio uses the matching original `AmpliCol` workload as its denominator, while every non-recurrence `pyAmpliCol` result is compared directly with the matching recurrence result at the stricter cross-mode tolerance. Built-in and UFO-SM recurrence results are likewise compared directly.

Runtime comparisons between these backends also include their vectorization strategy. The JIT rows can use SIMD across phase-space points, whereas the generated ASM and C++ rows are scalar. Their ratios therefore include both code-generation quality and vectorization; they are not comparisons of otherwise identical kernels.

Worked $d\bar{d} \rightarrow Zg$ example

For $n = 2$, the explicit process is

$$d(p_1) \bar{d}(p_2) \rightarrow Z(p_3) g(p_4).$$

The report labels source particles in process order. The selected leading-colour reference order for the selected-flow timing is therefore $(2, 4, 1, 3)$: the colour line enters through $\bar{d}$, traverses the gluon, and exits through d , while the colour-singlet Z remains in the public ordering only to keep source labels unambiguous. The corresponding coloured word is $(2, 4, 1)$.

`pyAmpliCol` builds a shared-current representation for the selected process. Source nodes carry particle, momentum, helicity, flavour, and colour-flow labels; composite nodes are identified by their complete quantum state. The selected-flow workload uses the traversal $(2, 4, 1)$ and sums helicities; the all-flow workload sums every LC flow for one selected helicity. Both layouts retain complete physical coverage and are validated against the same reference point before timing.

6.1 Built-in SM dedicated $d\bar{d} \rightarrow Z + \text{gluon}$ performance

Built-in SM $d\bar{d} \rightarrow Z + \text{gluon}$ ladder (block 1 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [$\mu\text{s}/\text{pt}$]	eval [$\mu\text{s}/\text{pt}$]	gen [s]	wall [$\mu\text{s}/\text{pt}$]	eval [$\mu\text{s}/\text{pt}$]
1 AmpliCol reference	2.13785	0.0218259	NOT EXPOSED	5.5e-05	0.0472668	NOT EXPOSED
1 compiled JIT O1	2.15616 (x1.01)	0.168534 (x7.72)	0.16852 (x7.72)	1.98119	0.0906541 (x1.92)	0.0906463 (x1.92)
1 compiled ASM O3	4.84424 (x2.27)	0.163975 (x7.51)	0.16396 (x7.51)	3.34914	0.0789603 (x1.67)	0.0789533 (x1.67)
1 compiled C++ O3	4.26369 (x1.99)	0.16242 (x7.44)	0.162404 (x7.44)	3.41819	0.0792272 (x1.68)	0.07922 (x1.68)
1 compiled JIT O3	2.26244 (x1.06)	0.175029 (x8.02)	0.175011 (x8.02)	1.97074	0.0876553 (x1.85)	0.0876473 (x1.85)
1 eager-DAG JIT O2	2.08816 (x0.977)	0.277385 (x12.7)	0.27736 (x12.7)	2.11466	0.177567 (x3.76)	0.177549 (x3.76)
1 recurrence JIT O2	4.32277 (x2.02)	0.165243 (x7.57)	0.165198 (x7.57)	4.18894	0.0958811 (x2.03)	0.0958503 (x2.03)
2 AmpliCol reference	1.95535	0.126864	NOT EXPOSED	8.5e-05	0.110985	NOT EXPOSED
2 compiled JIT O1	2.65554 (x1.36)	0.391506 (x3.09)	0.391464 (x3.09)	2.13476	0.157233 (x1.42)	0.157216 (x1.42)
2 compiled ASM O3	6.15367 (x3.15)	0.39943 (x3.15)	0.399391 (x3.15)	4.29844	0.132243 (x1.19)	0.13223 (x1.19)
2 compiled C++ O3	6.02846 (x3.08)	0.395057 (x3.11)	0.395011 (x3.11)	5.15126	0.13455 (x1.21)	0.134536 (x1.21)
2 compiled JIT O3	2.43599 (x1.25)	0.373928 (x2.95)	0.373891 (x2.95)	2.19259	0.156135 (x1.41)	0.156117 (x1.41)
2 eager-DAG JIT O2	2.27053 (x1.16)	0.719735 (x5.67)	0.719662 (x5.67)	2.1932	0.302384 (x2.72)	0.302354 (x2.72)
2 recurrence JIT O2	4.30432 (x2.2)	0.470483 (x3.71)	0.470323 (x3.71)	4.3378	0.26921 (x2.43)	0.269113 (x2.42)
3 AmpliCol reference	2.18544	0.461433	NOT EXPOSED	0.000263	0.327872	NOT EXPOSED
3 compiled JIT O1	3.51241 (x1.61)	0.735904 (x1.59)	0.735819 (x1.59)	2.71571	0.316683 (x0.966)	0.31664 (x0.966)
3 compiled ASM O3	14.0155 (x6.41)	1.11775 (x2.42)	1.11759 (x2.42)	7.75172	0.2639 (x0.805)	0.263869 (x0.805)
3 compiled C++ O3	13.584 (x6.22)	1.09653 (x2.38)	1.0964 (x2.38)	7.68248	0.254648 (x0.777)	0.254619 (x0.777)
3 compiled JIT O3	3.56644 (x1.63)	0.739976 (x1.6)	0.73988 (x1.6)	2.76149	0.312979 (x0.955)	0.312941 (x0.954)
3 eager-DAG JIT O2	2.18065 (x0.998)	1.52271 (x3.3)	1.52254 (x3.3)	2.25312	0.609896 (x1.86)	0.609826 (x1.86)
3 recurrence JIT O2	4.9238 (x2.25)	1.24476 (x2.7)	1.24419 (x2.7)	4.61935	0.843458 (x2.57)	0.843098 (x2.57)

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Built-in SM $d\bar{d} \rightarrow Z + \text{gluon ladder}$ (block 2 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [$\mu\text{s}/\text{pt}$]	eval [$\mu\text{s}/\text{pt}$]	gen [s]	wall [$\mu\text{s}/\text{pt}$]	eval [$\mu\text{s}/\text{pt}$]
4 AmpliCol reference	2.08378	1.46051	NOT EXPOSED	0.001244	1.18532	NOT EXPOSED
4 compiled JIT O1	6.35969 (x3.05)	2.04514 (x1.4)	2.04486 (x1.4)	4.58632	0.904782 (x0.763)	0.904664 (x0.763)
4 compiled ASM O3	36.2904 (x17.4)	4.81438 (x3.3)	4.81375 (x3.3)	24.1435	1.15552 (x0.975)	1.15537 (x0.975)
4 compiled C++ O3	38.6688 (x18.6)	4.33817 (x2.97)	4.33756 (x2.97)	21.2539	1.15908 (x0.978)	1.15893 (x0.978)
4 compiled JIT O3	6.28422 (x3.02)	2.06676 (x1.42)	2.06649 (x1.41)	4.78483	0.879554 (x0.742)	0.87944 (x0.742)
4 eager-DAG JIT O2	2.3523 (x1.13)	4.41129 (x3.02)	4.41075 (x3.02)	3.36306	2.10276 (x1.77)	2.1025 (x1.77)
4 recurrence JIT O2	4.81681 (x2.31)	3.0769 (x2.11)	3.07539 (x2.11)	4.33857	3.98523 (x3.36)	3.98341 (x3.36)
5 AmpliCol reference	2.33759	4.55676	NOT EXPOSED	0.009753	6.52985	NOT EXPOSED
5 compiled JIT O1	19.7547 (x8.45)	5.87459 (x1.29)	5.8737 (x1.29)	15.6925	4.76601 (x0.73)	4.76529 (x0.73)
5 compiled ASM O3	141.439 (x60.5)	19.4415 (x4.27)	19.4373 (x4.27)	101.382	13.8417 (x2.12)	13.8396 (x2.12)
5 compiled C++ O3	143.747 (x61.5)	18.4125 (x4.04)	18.4098 (x4.04)	100.349	13.7088 (x2.1)	13.7066 (x2.1)
5 compiled JIT O3	19.1461 (x8.19)	6.43862 (x1.41)	6.43767 (x1.41)	15.9056	4.88389 (x0.748)	4.88316 (x0.748)
5 eager-DAG JIT O2	3.31858 (x1.42)	13.308 (x2.92)	13.3058 (x2.92)	17.7414	12.0892 (x1.85)	12.0874 (x1.85)
5 recurrence JIT O2	5.54584 (x2.37)	6.39498 (x1.4)	6.39126 (x1.4)	4.893	8.32379 (x1.27)	8.31904 (x1.27)
6 AmpliCol reference	2.95044	13.6257	NOT EXPOSED	0.088403	41.4344	NOT EXPOSED
6 compiled JIT O1	113.05 (x38.3)	16.9407 (x1.24)	16.9372 (x1.24)	100.024	34.2265 (x0.826)	34.2205 (x0.826)
6 compiled ASM O3	966.611 (x328)	74.5663 (x5.47)	74.5509 (x5.47)	774.978	273.49 (x6.6)	273.435 (x6.6)
6 compiled C++ O3	970.673 (x329)	75.6452 (x5.55)	75.628 (x5.55)	785.254	253.271 (x6.11)	253.226 (x6.11)
6 compiled JIT O3	109.823 (x37.2)	18.3137 (x1.34)	18.3103 (x1.34)	99.5151	36.1164 (x0.872)	36.1102 (x0.872)
6 eager-DAG JIT O2	4.16545 (x1.41)	35.5061 (x2.61)	35.4988 (x2.61)	285.696	108.178 (x2.61)	108.158 (x2.61)
6 recurrence JIT O2	7.74161 (x2.62)	15.6393 (x1.15)	15.6288 (x1.15)	9.44746	50.1756 (x1.21)	50.1414 (x1.21)

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Built-in SM $d\bar{d} \rightarrow Z + \text{gluon ladder}$ (block 3 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [μ s/pt]	eval [μ s/pt]	gen [s]	wall [μ s/pt]	eval [μ s/pt]
7 AmpliCol reference	4.68311	38.9469	NOT EXPOSED	1.07804	323.294	NOT EXPOSED
7 compiled JIT O1	23.2629 (x4.97)	49.301 (x1.27)	49.2858 (x1.27)	1125.55	432.679 (x1.34)	432.575 (x1.34)
7 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
7 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
7 compiled JIT O3	23.282 (x4.97)	51.362 (x1.32)	51.3476 (x1.32)	1135.69	456.341 (x1.41)	456.24 (x1.41)
7 eager-DAG JIT O2	8.49475 (x1.81)	95.5278 (x2.45)	95.4996 (x2.45)	>15GB	>15GB	>15GB
7 recurrence JIT O2	15.1357 (x3.23)	39.4808 (x1.01)	39.4341 (x1.01)	59.1777	493.994 (x1.53)	493.484 (x1.53)
8 AmpliCol reference	8.94027	95.7623	NOT EXPOSED	13.2762	6178.12	NOT EXPOSED
8 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
8 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
8 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
8 compiled JIT O3	62.6861 (x7.01)	157.219 (x1.64)	157.133 (x1.64)	>15GB	>15GB	>15GB
8 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
8 recurrence JIT O2	63.3102 (x7.08)	93.1528 (x0.973)	92.9231 (x0.97)	1097.31	8440.19 (x1.37)	8436.74 (x1.37)
9 AmpliCol reference	26.0637	227.834	NOT EXPOSED	>15GB	>15GB	NOT EXPOSED
9 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
9 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
9 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
9 compiled JIT O3	>15GB	>15GB	>15GB	BLOCKED DEP.	BLOCKED DEP.	BLOCKED DEP.
9 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
9 recurrence JIT O2	758.197 (x29.1)	235.764 (x1.03)	232.585 (x1.02)	>1h	>1h	>1h

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6.2 UFO-SM dedicated $d\bar{d} \rightarrow Z + \text{gluon}$ performance

UFO-SM $d\bar{d} \rightarrow Z + \text{gluon}$ ladder (block 1 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [μ s/pt]	eval [μ s/pt]	gen [s]	wall [μ s/pt]	eval [μ s/pt]
1 AmpliCol reference	2.13785	0.0218259	NOT EXPOSED	5.5e-05	0.0472668	NOT EXPOSED
1 compiled JIT O1	3.04436 (x1.42)	0.176466 (x8.09)	0.17645 (x8.08)	3.22026	0.0926795 (x1.96)	0.0926712 (x1.96)
1 compiled ASM O3	6.79119 (x3.18)	0.166489 (x7.63)	0.166475 (x7.63)	4.95337	0.086196 (x1.82)	0.086188 (x1.82)
1 compiled C++ O3	6.59103 (x3.08)	0.17618 (x8.07)	0.176165 (x8.07)	5.1468	0.0827418 (x1.75)	0.0827347 (x1.75)
1 compiled JIT O3	3.16759 (x1.48)	0.173083 (x7.93)	0.173068 (x7.93)	3.02743	0.0884816 (x1.87)	0.088474 (x1.87)
1 eager-DAG JIT O2	3.02416 (x1.41)	0.285784 (x13.1)	0.285758 (x13.1)	3.20005	0.177485 (x3.75)	0.17747 (x3.75)
1 recurrence JIT O2	7.77959 (x3.64)	0.214993 (x9.85)	0.214932 (x9.85)	8.07008	0.0989993 (x2.09)	0.0989649 (x2.09)
2 AmpliCol reference	1.95535	0.126864	NOT EXPOSED	8.5e-05	0.110985	NOT EXPOSED
2 compiled JIT O1	3.72184 (x1.9)	0.40191 (x3.17)	0.401869 (x3.17)	3.05346	0.167961 (x1.51)	0.167942 (x1.51)
2 compiled ASM O3	8.40786 (x4.3)	0.430985 (x3.4)	0.43094 (x3.4)	6.54365	0.134142 (x1.21)	0.134128 (x1.21)
2 compiled C++ O3	8.86995 (x4.54)	0.434686 (x3.43)	0.434629 (x3.43)	6.24065	0.137264 (x1.24)	0.13725 (x1.24)
2 compiled JIT O3	3.39839 (x1.74)	0.397734 (x3.14)	0.397692 (x3.13)	3.28912	0.16711 (x1.51)	0.167092 (x1.51)
2 eager-DAG JIT O2	3.35166 (x1.71)	0.741497 (x5.84)	0.741426 (x5.84)	3.37404	0.308342 (x2.78)	0.30831 (x2.78)
2 recurrence JIT O2	7.4159 (x3.79)	0.53512 (x4.22)	0.534939 (x4.22)	7.34434	0.300511 (x2.71)	0.300387 (x2.71)
3 AmpliCol reference	2.18544	0.461433	NOT EXPOSED	0.000263	0.327872	NOT EXPOSED
3 compiled JIT O1	4.47321 (x2.05)	0.803056 (x1.74)	0.802962 (x1.74)	3.53612	0.383198 (x1.17)	0.383158 (x1.17)
3 compiled ASM O3	17.0605 (x7.81)	1.1665 (x2.53)	1.16636 (x2.53)	9.45832	0.260667 (x0.795)	0.260636 (x0.795)
3 compiled C++ O3	16.6795 (x7.63)	1.17009 (x2.54)	1.16994 (x2.54)	9.33315	0.261195 (x0.797)	0.261163 (x0.797)
3 compiled JIT O3	4.54024 (x2.08)	0.798658 (x1.73)	0.798562 (x1.73)	3.64368	0.31469 (x0.96)	0.314653 (x0.96)
3 eager-DAG JIT O2	3.22658 (x1.48)	1.58168 (x3.43)	1.58147 (x3.43)	3.28794	0.620922 (x1.89)	0.620856 (x1.89)
3 recurrence JIT O2	7.60922 (x3.48)	1.33478 (x2.89)	1.33421 (x2.89)	7.85757	0.911843 (x2.78)	0.911423 (x2.78)

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UFO-SM $d\bar{d} \rightarrow Z + \text{gluon ladder}$ (block 2 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [$\mu\text{s}/\text{pt}$]	eval [$\mu\text{s}/\text{pt}$]	gen [s]	wall [$\mu\text{s}/\text{pt}$]	eval [$\mu\text{s}/\text{pt}$]
4 AmpliCol reference	2.08378	1.46051	NOT EXPOSED	0.001244	1.18532	NOT EXPOSED
4 compiled JIT O1	7.7462 (x3.72)	2.25221 (x1.54)	2.25192 (x1.54)	5.56249	0.930108 (x0.785)	0.929993 (x0.785)
4 compiled ASM O3	39.0533 (x18.7)	4.40397 (x3.02)	4.4034 (x3.01)	23.051	1.1714 (x0.988)	1.17125 (x0.988)
4 compiled C++ O3	38.681 (x18.6)	4.42642 (x3.03)	4.42579 (x3.03)	23.1907	1.1721 (x0.989)	1.17194 (x0.989)
4 compiled JIT O3	7.54326 (x3.62)	2.34951 (x1.61)	2.34921 (x1.61)	5.7513	0.940321 (x0.793)	0.940204 (x0.793)
4 eager-DAG JIT O2	3.39852 (x1.63)	4.47225 (x3.06)	4.47169 (x3.06)	4.52215	2.12884 (x1.8)	2.12856 (x1.8)
4 recurrence JIT O2	7.97042 (x3.82)	3.38685 (x2.32)	3.38519 (x2.32)	7.51221	3.70527 (x3.13)	3.70368 (x3.12)
5 AmpliCol reference	2.33759	4.55676	NOT EXPOSED	0.009753	6.52985	NOT EXPOSED
5 compiled JIT O1	21.2448 (x9.09)	6.70989 (x1.47)	6.70886 (x1.47)	17.6205	4.98559 (x0.764)	4.98461 (x0.763)
5 compiled ASM O3	155.192 (x66.4)	18.4454 (x4.05)	18.4418 (x4.05)	102.596	14.0671 (x2.15)	14.065 (x2.15)
5 compiled C++ O3	149.668 (x64)	18.4372 (x4.05)	18.4343 (x4.05)	101.733	14.0079 (x2.15)	14.0058 (x2.14)
5 compiled JIT O3	21.4625 (x9.18)	7.28371 (x1.6)	7.28253 (x1.6)	17.9595	5.25702 (x0.805)	5.25622 (x0.805)
5 eager-DAG JIT O2	4.06502 (x1.74)	12.8182 (x2.81)	12.8161 (x2.81)	19.7055	11.957 (x1.83)	11.955 (x1.83)
5 recurrence JIT O2	8.91905 (x3.82)	6.89283 (x1.51)	6.88896 (x1.51)	8.22631	8.53054 (x1.31)	8.52565 (x1.31)
6 AmpliCol reference	2.95044	13.6257	NOT EXPOSED	0.088403	41.4344	NOT EXPOSED
6 compiled JIT O1	116.641 (x39.5)	19.6218 (x1.44)	19.6181 (x1.44)	104.453	37.844 (x0.913)	37.8347 (x0.913)
6 compiled ASM O3	990.116 (x336)	84.0434 (x6.17)	84.0282 (x6.17)	847.661	229.167 (x5.53)	229.119 (x5.53)
6 compiled C++ O3	1001.81 (x340)	93.3157 (x6.85)	93.2989 (x6.85)	858.972	231.285 (x5.58)	231.243 (x5.58)
6 compiled JIT O3	115.145 (x39)	22.3596 (x1.64)	22.3554 (x1.64)	116.56	49.4569 (x1.19)	49.4434 (x1.19)
6 eager-DAG JIT O2	5.7651 (x1.95)	34.4271 (x2.53)	34.4204 (x2.53)	293.243	102.462 (x2.47)	102.445 (x2.47)
6 recurrence JIT O2	11.353 (x3.85)	17.0621 (x1.25)	17.05 (x1.25)	12.8688	51.1371 (x1.23)	51.1034 (x1.23)

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UFO-SM $d\bar{d} \rightarrow Z + \text{gluon ladder}$ (block 3 of 3)

n setup	selected flow, helicity sum			all flows, single helicity		
	gen [s]	wall [μ s/pt]	eval [μ s/pt]	gen [s]	wall [μ s/pt]	eval [μ s/pt]
7 AmpliCol reference	4.68311	38.9469	NOT EXPOSED	1.07804	323.294	NOT EXPOSED
7 compiled JIT O1	25.4179 (x5.43)	55.6742 (x1.43)	55.6587 (x1.43)	1315.65	491.428 (x1.52)	491.312 (x1.52)
7 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
7 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
7 compiled JIT O3	26.0181 (x5.56)	64.3771 (x1.65)	64.3568 (x1.65)	1304.39	508.598 (x1.57)	508.465 (x1.57)
7 eager-DAG JIT O2	9.143 (x1.95)	88.6295 (x2.28)	88.6045 (x2.28)	>15GB	>15GB	>15GB
7 recurrence JIT O2	19.4086 (x4.14)	41.4298 (x1.06)	41.3769 (x1.06)	66.4673	525.93 (x1.63)	525.372 (x1.63)
8 AmpliCol reference	8.94027	95.7623	NOT EXPOSED	13.2762	6178.12	NOT EXPOSED
8 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
8 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
8 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
8 compiled JIT O3	70.5997 (x7.9)	181.081 (x1.89)	180.962 (x1.89)	>15GB	>15GB	>15GB
8 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
8 recurrence JIT O2	71.439 (x7.99)	98.6816 (x1.03)	98.416 (x1.03)	1164	8854.14 (x1.43)	8850.37 (x1.43)
9 AmpliCol reference	26.0637	227.834	NOT EXPOSED	>15GB	>15GB	NOT EXPOSED
9 compiled JIT O1	N/A	N/A	N/A	N/A	N/A	N/A
9 compiled ASM O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
9 compiled C++ O3	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A	STATIC N/A
9 compiled JIT O3	>15GB	>15GB	>15GB	BLOCKED DEP.	BLOCKED DEP.	BLOCKED DEP.
9 eager-DAG JIT O2	N/A	N/A	N/A	N/A	N/A	N/A
9 recurrence JIT O2	811.911 (x31.2)	241.255 (x1.06)	238.949 (x1.05)	BLOCKED DEP.	BLOCKED DEP.	BLOCKED DEP.

Original AmpliCol is the denominator only when successful and exactly linked by stored same-point evidence, directly or through current recurrence. Otherwise its status remains visible and successful pyAmpliCol values are absolute. Unverified diagnostics retain absolute clocks with a yellow marker, but never enter ratios or summaries. Each pyAmpliCol row reports separate topology-replay and all-flow-union generation and runtime measurements. Parenthesized values are candidate/reference ratios. All-flow generation shows the absolute pyAmpliCol value without a ratio because its setup boundary differs from the reference. The wall time is the common runtime observable. Not exposed means that a successful wall measurement has no dedicated authenticated evaluator-total timing. Authenticated accumulated warmed evaluator totals appear in the eval column with enough precision to remain distinct from wall time. Its parenthesized multiplier uses the original-AmpliCol wall measurement as denominator because the legacy reference has no separate evaluator-total clock. Recurrence rows retain the independently measured narrower recurrence core in detailed evidence; it is not relabeled as evaluator total. Neither evaluator total nor recurrence core is derived from wall time or from the other. Older entries without authenticated total evidence remain marked not exposed; it is not a missing measurement. user cap: native C++/ASM generation is not attempted above n=6 (native-backend-generation-cap-n6-v1).

Each LC workload reports process-generation time in seconds and native wall time in microseconds per phase-space point, together with a separately measured execution attribution when exposed. Compiled rows use the labelled JIT, ASM, or C++ backend; eager and recurrence rows use portable JIT O2 prepared models. Parenthesized wall-time and selected-flow generation values compare each pyAmpliCol row with the corresponding AmpliCol workload. For compiled and eager rows, wall time is the authenticated warmed total-evaluator boundary; not exposed refers only to a narrower execution attribution. All-flow generation is shown absolutely without a ratio because its setup boundary differs from the reference.

7 Colour-Singlet Model Ladders

Both colour-singlet ladders use compiled JIT O3 evaluation and the full-colour contract. The scalar-contact ladder retains $n = 2, \dots, 8$ final scalars and tests high-valence contact decomposition, identical-particle normalization, and source parity without Standard-Model-specific kernels. The scalar-gravity ladder retains $n = 2, \dots, 4$ final gravitons and exercises spin-two wavefunctions, tensor propagators, and momentum-dependent vertices. Because every external particle in these models is a colour singlet, LC, NLC, and full colour reduce to the same unit colour contraction.

For these ladders, the relative-difference row is $||\mathcal{M}|_{f64}^2 - |\mathcal{M}|_{hp}^2| / \max(|\mathcal{M}|_{hp}^2, 10^{-300})$, where the high-precision value uses the same phase-space point. It is a numerical stability check, not a performance ratio.

7.1 Scalar-contact ladder

metric	scalar_0 scalar_0 > n*scalar_0						
	n=2	n=3	n=4	n=5	n=6	n=7	n=8
generation [s]	1.46882	1.70434	2.7394	6.19946	19.244	64.393	N/A
wall [μ s/pt]	0.102384	0.143562	0.33282	0.537016	1.44194	2.01395	N/A
evaluator total [μ s/pt]	0.102375	0.143548	0.33278	0.536947	1.44173	2.01363	N/A
matrix element	0.5	0.167	0.0417	0.00833	0.00139	0.000198	N/A
relative diff. vs hp	1.11e-16	1.67e-16	0	0	0	0	N/A

Wall time is the common runtime observable. Evaluator total is the independently measured accumulated warmed evaluator clock from its dedicated authenticated record; it is never copied from or derived from wall time. Unverified diagnostics retain absolute generation, wall, and evaluator clocks with a yellow marker; they are not successes. NOT EXPOSED denotes a successful wall measurement, and older entries without dedicated evaluator-total evidence remain valid; it does not denote a missing result.

7.2 Scalar-gravity ladder

metric	scalar_0 scalar_0 > n*graviton		
	n=2	n=3	n=4
generation [s]	2.2086	185.114	1173.09
wall [μ s/pt]	2.19559	1509.22	35128.9
evaluator total [μ s/pt]	2.1954	1509.08	35127.9
matrix element	1.61e+09	9.46e+11	3.07e+13
relative diff. vs hp	1.04e-15	4.13e-15	1.91e-14

Wall time is the common runtime observable. Evaluator total is the independently measured accumulated warmed evaluator clock from its dedicated authenticated record; it is never copied from or derived from wall time. Unverified diagnostics retain absolute generation, wall, and evaluator clocks with a yellow marker; they are not successes. NOT EXPOSED denotes a successful wall measurement, and older entries without dedicated evaluator-total evidence remain valid; it does not denote a missing result.

8 Interpretation

Performance comparisons are meaningful only for a baseline measured with the same observables, parameters, selectors, and hardware profile. The result tables therefore present validated, architecture-local comparisons; they do not combine measurements from different machines. An N/A entry denotes only a structurally inapplicable cell or observable; every applicable cell contains a validated measurement or documented resource-limit status. Numerical agreement, timing uncertainty, and software revision remain part of the published result set even when a compact table shows only the central timing.

9 Worked $d\bar{d} \rightarrow Zgg$ Example

This example illustrates how a familiar tree-level process is represented by reusable off-shell currents. It also fixes the momentum, helicity, and colour conventions used in the performance tables.

9.1 External legs and conventions

The physical process is

$$d(p_1) \bar{d}(p_2) \longrightarrow Z(p_3) g(p_4) g(p_5), \quad p_1 + p_2 = p_3 + p_4 + p_5. \quad (6)$$

The current recursion uses the all-outgoing convention

$$q_1 = -p_1, \quad q_2 = -p_2, \quad q_i = p_i \quad (i = 3, 4, 5), \quad \sum_{i=1}^5 q_i = 0. \quad (7)$$

External momenta are supplied in the physical convention of Eq. (6) and crossed to the all-outgoing convention before the recursion is evaluated.

label	physical leg	outgoing state	momentum	states	colour role
1	incoming $\bar{d}$	outgoing $\bar{d}$	$q_1 = -p_1$	2	$\bar{\mathbf{3}}$, line end
2	incoming d	outgoing d	$q_2 = -p_2$	2	$\mathbf{3}$, line start
3	outgoing Z	outgoing Z	$q_3 = p_3$	3	singlet
4	outgoing g	outgoing g	$q_4 = p_4$	2	adjoint insertion
5	outgoing g	outgoing g	$q_5 = p_5$	2	adjoint insertion

There are $2 \times 2 \times 3 \times 2 \times 2 = 48$ physical helicity configurations. Twenty-four vanish identically because the massless quark line permits only chirality-compatible initial-state helicities; they remain explicit zero entries in resolved results.

At leading colour the Z is absent from the colour word. Complete coverage contains the two orderings

$$c_0 = (2, 4, 5, 1), \quad c_1 = (2, 5, 4, 1). \quad (8)$$

Both are retained during generation, so either flow can be selected at evaluation time.

9.2 Reusable current recursion

For a nonempty source subset I , define its momentum by

$$q_I = \sum_{i \in I} q_i. \quad (9)$$

A one-leg current is the appropriate external spinor, polarization vector, or scalar wavefunction. Composite currents obey the binary recursion

$$J_{t,\Omega}^a(I) = P_t^a{}_b(q_I) \sum_{\substack{I=A \dot{\cup} B \\ A, B \neq \emptyset}} \sum_{v: t_A t_B \rightarrow t} \kappa_v V_v^b{}_{cd}(q_A, q_B) J_{t_A, \Omega_A}^c(A) J_{t_B, \Omega_B}^d(B). \quad (10)$$

Here t is the particle state, P_t is its propagator, V_v is an allowed interaction, and Ω collects the quantum numbers required to identify the current. Only combinations consistent with the model, perturbative order, and colour ordering enter the sum.

Representative terms along the quark line are

$$J_d(2, 5) = P_d V_{dgd} [J_d(2), J_g(5)], \quad (11)$$

$$J_g(4, 5) = P_g V_{ggg} [J_g(4), J_g(5)], \quad (12)$$

$$J_d(2, 4, 5) = P_d \left(V_{dgd} [J_d(2, 5), J_g(4)] + V_{dgd} [J_d(2, 4), J_g(5)] + V_{dgd} [J_d(2), J_g(4, 5)] \right), \quad (13)$$

$$J_d(2, 3, 4, 5) = P_d \sum_{A \dot{\cup} B = \{2, 3, 4, 5\}} V [J(A), J(B)]. \quad (14)$$

Equation (14) includes every admissible placement of the Z emission and both gluon insertions. Ordered currents such as $J_d(2, 5, 4)$ and $J_d(2, 4, 5)$ remain distinct even though their momentum subset is the same.

The final closure joins the endpoint current to source label 1:

$$\mathcal{A}_{h,c}^{(r)} = C_{\dot{a}a}^{(r)} J_{\dot{a}}^a(1) J_d^a(2, 3, 4, 5), \quad \mathcal{A}_{h,c} = \sum_{r \in G(h,c)} \mathcal{A}_{h,c}^{(r)}. \quad (15)$$

The set $G(h, c)$ contains contributions that must be summed coherently. Identical subcurrents are evaluated once and reused wherever their complete quantum-number and ordering labels agree. This sharing is the essential difference between the current DAG and an independent evaluation of every Feynman diagram.

9.3 Resolved observable and normalization

The two leading-colour layouts described in Sec. 11 evaluate the same complete set of helicity and flow components. One is optimized for a selected flow with summed helicities; the other shares work across all flows for a selected helicity. Their resolved components and their summed matrix element are numerically identical.

For the built-in Standard Model, the normalization used in this example is

$$|\mathcal{M}|^2 = \mathcal{N} \sum_{h,c} w_{h,c} \left| \sum_{r \in G(h,c)} \mathcal{A}_{h,c}^{(r)} \right|^2, \quad \mathcal{N} = \frac{27 (4\pi\alpha_s)^2 (8\pi\alpha_{EW})}{36 \times 2}. \quad (16)$$

The denominator averages over the incoming colour and spin states, while the factor $2 = 2!$ accounts for the identical final-state gluons. Coupling powers, colour weights, coherent groups, and symmetry factors are retained separately so that changing numerical model parameters does not alter the process definition.

10 Shared-Current DAG and Numerical Execution

The current recursion becomes a directed acyclic graph (DAG) when physically identical intermediate currents are shared. This section summarizes the identity and ordering needed for that reuse and relates them to the three execution modes.

10.1 Current identity and reuse

Two contributions share a current only when every physics-relevant label agrees. Schematically,

$$\Omega(J) = (t, I, H, s, F, Q, C, O, a, \pi_I), \quad (17)$$

where the entries identify the particle species, external subset, helicity and spin ancestry, flavour and additive charges, colour state, perturbative order, auxiliary-current type, and colour-sensitive ordering. The subset determines the current momentum $q_I = \sum_{i \in I} q_i$, while the ordered labels prevent distinct colour traversals from being merged.

Exact identity permits one stored current to serve every amplitude contribution that needs it. A second kind of reuse applies when different currents require the same numerical kernel value up to a known exact factor. Neither form of reuse merges distinct physical helicity or colour-flow components.

10.2 Topological ordering

The graph is evaluated in increasing external-subset size. Every input is therefore available before the current that consumes it. For the process in Sec. 9, the ordering is schematically

$$\boxed{\text{sources}} \longrightarrow \boxed{|I| = 2} \longrightarrow \boxed{|I| = 3} \longrightarrow \boxed{|I| = 4} \longrightarrow \boxed{\text{amplitude closures}}. \quad (18)$$

At each stage, all admissible partitions contributing to the same current are summed coherently before propagation. The final closures join the largest retained currents to the remaining external states. This ordering makes the dependency structure explicit without assigning physical meaning to a particular storage layout.

10.3 Numerical realization

Compiled mode combines many relations in a stage into process-specific multi-output evaluators. Eager mode follows the same process graph using reusable prepared-model kernels. Recurrence mode represents the current and closure sequence directly. All three preserve the same coherent sums, colour contraction, helicity treatment, symmetry weights, and normalization.

For resolved leading-colour evaluation, the public result has the form

$$R_{p,h,f} = |\mathcal{M}_{h,f}(p)|^2, \quad \text{shape}(R) = (N_{\text{point}}, N_{\text{helicity}}, N_{\text{flow}}). \quad (19)$$

Structural zeros remain explicit, and summing the non-point axes must reproduce the directly evaluated total within the stated numerical tolerance.

10.4 Timing boundary

The primary runtime quantity is the complete native wall time after caller-language array conversion. It includes source and momentum preparation, the mode-specific numerical evaluation, selectors, closures, and final reduction. Packing performed by the caller is outside this boundary.

Some backends additionally expose a narrower execution attribution. Because its precise boundary is mode-specific, it is reported only as a supplementary quantity and only when measured directly. The complete wall time remains the common observable used for comparisons; an unavailable attribution is marked explicitly rather than inferred.

11 Reusable Leading-Colour Execution Layouts

Leading-colour coverage and its numerical layout are separate concepts. Let $\mathcal{H}$ be the retained helicities and $\mathcal{F}$ the retained physical colour flows. Complete leading-colour coverage represents

$$\mathcal{C}_{\text{LC}} = \mathcal{H} \times \mathcal{F} \quad (20)$$

and can expose every resolved amplitude $\mathcal{A}_{hf}$. For selected subsets $S_H \subseteq \mathcal{H}$ and $S_F \subseteq \mathcal{F}$, the observable is

$$\mathcal{W}(S_H, S_F) = \mathcal{N} \sum_{h \in S_H} \sum_{f \in S_F} w_{hf} \left| \sum_{r \in G(h,f)} \mathcal{A}_{hf}^{(r)} \right|^2. \quad (21)$$

Here $G(h, f)$ is a coherent contribution group, w_{hf} contains exact symmetry and leading-colour weights, and $\mathcal{N}$ is the process normalization. The layout changes how this sum is scheduled, not its components or normalization.

11.1 Two complete-coverage layouts

The report measures two layouts because selected-flow and all-flow workloads offer different opportunities for sharing:

property	topology replay	all-flow union
Measured workload	one selected flow, summed helicities	all flows, one selected helicity
Retained physics	every flow and helicity	every flow and helicity
Main reuse	equivalent flow topologies related by exact permutations and factors	currents and closures shared directly across all flows
Availability	compiled, eager, and recurrence	compiled, eager, and recurrence

In topology replay, exact relations partition physical flows into equivalence classes. If $\rho(f)$ represents the class containing f , the calculation records a permutation π_f , an exact factor s_f , and the corresponding weight:

$$\mathcal{A}_{hf}(p) = s_f \mathcal{A}_{\pi_f(h), \rho(f)}(\pi_f(p)). \quad (22)$$

This avoids constructing an independent current graph for every physical ordering and is suited to selected-flow, helicity-summed evaluation.

The all-flow union instead merges the dependency graphs of all resolved components:

$$D_{\text{union}} = \left(\bigcup_{h \in \mathcal{H}, f \in \mathcal{F}} D_{hf} \right) / \sim, \quad (23)$$

where $\sim$ identifies currents, kernel values, and closures that are algebraically equal up to retained exact factors. For a selected helicity, one graph traversal then reuses intermediate currents across all flows.

Compiled mode specializes the chosen layout to process-wide stage evaluators. Eager mode applies prepared-model kernels to the same dependency structure, and recurrence mode constructs a compact current schedule directly. These execution choices do not change $\mathcal{H}$, $\mathcal{F}$, or Eq. (21).

11.2 Runtime selection

Flow and helicity selections refer to physical components. A selection may be applied to a complete batch or specified independently for each point; omitting an axis sums all retained entries on that axis. Rows with equal selections can be grouped into contiguous numerical batches. This grouping changes neither the generated coverage nor the resolved result.

11.3 Generation and reference boundaries

The two layouts build different graphs and numerical schedules, so their process-generation times are distinct observables:

$$T_{\text{gen}}^{\text{replay}} \quad \text{and} \quad T_{\text{gen}}^{\text{union}}. \quad (24)$$

Cross-mode comparisons use the same layout on both sides. Preparation of reusable model-level kernels occurs before the process-generation timer and is not included in these table values.

The original-**AmpliCol** reference uses two different routes. For the selected-flow workload, its generation value measures creation of the process library. For the all-flow workload, the reported setup value measures preparation of direct all-flow evaluation. The corresponding runtime measurements implement the same physical workloads as **pyAmpliCol**, but the all-flow setup quantities do not define a like-for-like process-generation speedup. The table therefore reports both quantities absolutely and marks their generation comparison as not comparable.

11.4 NLC and full colour

The alternative layouts apply only at leading colour. NLC and full-colour evaluation construct an amplitude basis and contract it with the appropriate colour metric:

$$\mathcal{W}_{\text{contracted}} = \mathcal{N} \sum_h \sum_{a,b} \mathcal{A}_{ha}^* C_{ab} \mathcal{A}_{hb}. \quad (25)$$

These treatments do not expose the physical leading-colour flow axis, so there is no topology-replay or all-flow-union distinction.

12 Compiled, Eager, and Recurrence Execution

The three execution modes differ in how they turn the same current recursion into numerical work. Compiled mode specializes groups of current relations to one process. Eager mode applies reusable model kernels according to a process-specific schedule. Recurrence mode constructs a compact schedule of currents, contributions, and closures directly. These choices change the balance between process-generation cost and runtime work, but not the observable being evaluated.

Recurrence JIT O2 is the default execution mode.

mode	process preparation	evaluation	characteristic balance
Compiled	Builds process-specific stage evaluators	Evaluates a small number of process-wide, multi-output stages	More preparation; greater fusion across a process
Eager	Maps the process graph to reusable prepared-model kernels	Applies those kernels according to the process dependencies	Reuses model work across processes
Recurrence	Builds a compact schedule directly from the current recursion	Evaluates scheduled currents, contributions, and closures	Avoids construction of a generic process DAG

12.1 Common physics contract

All three modes retain the same external states, perturbative orders, helicities, colour components, coherent amplitude groups, symmetry factors, and initial-state averages. In a contracted colour basis their common observable has the schematic form

$$\mathcal{M}_p = \mathcal{N} \sum_h \sum_{a,b} A_{ha}(p)^* C_{ab} A_{hb}(p), \quad (26)$$

with the corresponding resolved form retained when helicity or leading-colour flow components are requested. Each mode must reproduce both the directly evaluated total and the sum of those resolved components.

Current contributions are accumulated coherently before a propagator or other current transformation is applied. Final closures then form the amplitude components entering Eq. (26). This ordering is common to the three modes and prevents execution strategy from changing the normalization.

12.2 Prepared-model boundary

A prepared model contains reusable numerical information derived from the model’s particles, parameters, propagators, and interaction tensors. Eager and recurrence JIT O2 measurements reuse this model-level preparation and add only the process-dependent schedule. Compiled JIT O3 measurements instead include construction and compilation of their process-specific evaluators.

Prepared-model creation occurs outside the process-generation timer. The reported eager and recurrence generation times therefore describe adding a process to an already prepared model, not importing and compiling the model itself.

12.3 Batches, selectors, and portability

Evaluation is batched over phase-space points, allowing the JIT backend to use the point dimension for vectorization. Partitioning a large input batch changes neither the process representation nor the numerical result.

Helicity and colour-flow selectors refer to physical resolved components. A complete leading-colour calculation retains every declared helicity and flow, so selecting one flow with summed helicities or one helicity with all flows does not alter the underlying physics coverage. The two layouts used for these workloads are described in Sec. 11.

Portable JIT state is compiled for the receiving processor when loaded. Target-native ASM and C++ payloads remain architecture-specific, and the exact symbolic representation is retained where higher-precision evaluation is supported.

12.4 Interpretation

Compiled mode may benefit most when process-wide fusion outweighs its additional preparation cost. Eager mode can reduce repeated preparation when many processes share a model, while recurrence provides a compact direct representation of the current construction. The tables compare these trade-offs under identical physics, selector, precision, and validation conditions; no mode is assumed to be universally fastest.

13 Supported UFO and Serialized-JSON Models

pyAmpliCol accepts a documented subset of the Universal FeynRules Output (UFO) format and an equivalent serialized-JSON representation. Both sources are normalized before process generation and then use the same current recursion, colour treatment, and numerical interfaces as the built-in models. Unsupported structures are rejected explicitly rather than interpreted approximately.

13.1 Sources and normalization

The two input routes are

$$\{\text{trusted UFO module, serialized JSON}\} \longrightarrow \text{normalized model} \longrightarrow \text{numerical kernels.} \quad (27)$$

A UFO module is executable Python and must therefore come from a trusted source. The serialized JSON form is data-only and is preferable for portable exchange and archival. After loading, both forms describe the same particles, parameters, propagators, interactions, coupling orders, Lorentz tensors, and colour tensors.

Model symbols remain in a model-local namespace, so equally named parameters from different models are not identified accidentally. Normalization also makes component orderings and particle conjugation explicit. A model is accepted for ordinary process generation only when the normalized model satisfies the support boundary below; an unrepresentable requested interaction is rejected explicitly.

13.2 Interactions and exact verification

A UFO vertex is a sum over colour and Lorentz structures,

$$\mathcal{V}(1, \dots, n) = \sum_{a,b} g_{ab} C_a(c_1, \dots, c_n) L_b(s_1, p_1; \dots; s_n, p_n). \quad (28)$$

Each term retains its coupling-order assignment. The supported Lorentz basis includes scalars, momenta, metrics, Dirac identity and gamma matrices, γ_5 , chiral projectors, slashed momenta, and sigma tensors. The supported SU(3) representations are the singlet, fundamental, antifundamental, and adjoint.

Colour projection and symmetry reuse are based on exact normalized expressions. Numerical agreement at a sample point is not used to establish an algebraic identity. When an identity needed only for reuse cannot be proved, the corresponding contributions are evaluated separately. When an identity is required to represent an interaction at all, failure to establish it causes the model or process to be rejected.

Leading-colour singlet subtraction, for example, follows from the fundamental Fierz identity

$$(T^a)_i^{\bar{j}} (T^a)_k^{\bar{l}} = T_R \left(\delta_i^{\bar{l}} \delta_k^{\bar{j}} - \frac{1}{N_c} \delta_i^{\bar{j}} \delta_k^{\bar{l}} \right). \quad (29)$$

It is used only when the colour representation, Lorentz interaction, and vector propagator satisfy the corresponding exact conditions. The same principle applies to vectorlike parity, Yang–Mills relations, and current reflection.

13.3 Contact interactions and propagators

Supported contact interactions are rewritten with non-propagating auxiliary currents only when the original tensor can be reconstructed exactly. The physical coupling appears once, and the rewriting introduces no artificial pole. Colour-singlet scalar contacts provide the validation ladder for this construction. Higher-valence singlet interactions are accepted when their normalized tensor admits the same exact reconstruction.

A coloured four-vector gauge contact is accepted only for the two- f structure

$$C^{a_1 a_2 a_3 a_4} = c f^{a_i a_j b} f^{b a_k a_r}, \quad (30)$$

up to exact permutations, with one shared adjoint index and a matching Lorentz factorization. Other coloured contact tensors require a separately supported intermediate colour channel.

Standard scalar, Dirac, vector, and spin-two propagators use the mass, width, gauge, and component ordering declared by the normalized model. A custom UFO propagator is accepted only when its numerator and denominator can be lowered exactly; optimizations that assume a standard propagator are then disabled. Massless spin two is validated by the scalar-gravity model. Massive spin-two support remains experimental because it has not yet been tested across a comparably broad set of models.

13.4 Support boundary

In the table, *conditional* means that the stated exact algebraic condition must be established for the particular model expression. It does not imply acceptance of every algebraically equivalent UFO spelling.

area	class	status	boundary
source	UFO directory / JSON	supported	UFO Python must be trusted; JSON is data-only and both use the same normalized model

area	class	status	boundary
spin	scalar, Dirac, vector	supported	standard wavefunctions, propagators, component orderings, and closures
spin	massless spin two	supported	tensor path validated by the scalar-gravity model
spin	massive spin two	experimental	Fierz–Pauli path is available, but broad model validation is incomplete
spin	UFO ghosts	partial	propagating ghosts require an exactly representable custom propagator and are not broadly qualified
spin	Majorana/FNV, spin 3/2	unsupported	no fermion-number-violating-flow or Rarita–Schwinger current prescription
colour	1, 3, $\bar{3}$, 8 of SU(3)	supported	LC, NLC, and full-colour treatments described in this report
colour	6, $\bar{6}$, extra groups	unsupported	no sextet or multiple-group flow and contraction representation
arity 3	1, δ , T , f colour	supported	exact colour projection and representable Lorentz components
arity 3	d^{abc} , ϵ , K_6 , T_6	unsupported	no supported colour-flow and contraction prescription
arity 4	colour singlet	conditional	exact tensor reconstruction by non-propagating auxiliary currents
arity 4	two- f gauge colour	conditional	one shared adjoint index, trivalent topology, and exact Lorentz reconstruction
arity 4	other coloured tensors	unsupported	no supported auxiliary colour channel
arity > 4	colour-singlet contact	conditional	exact no-pole reconstruction with the original coupling inserted once
arity > 4	coloured contact	unsupported	no general irreducible-channel factorization
Lorentz	normalized tensor basis	supported	every tensor function must normalize with explicit component axes
Lorentz	form factors / unknown functions	unsupported	no portable numerical prescription
propagator	standard 0, $\frac{1}{2}$, 1, 2	supported	model mass and width, with explicit component ordering and closure
propagator	custom UFO tensor	conditional	exact component lowering; standard-propagator optimizations are disabled
symmetry	vectorlike/Yang–Mills/reflection	conditional	exact normalized identities over the complete interaction sector used by the process

References

- [1] F. A. Berends and W. T. Giele, *Recursive Calculations for Processes with n Gluons*, Nucl. Phys. B **306** (1988) 759–808, doi:10.1016/0550-3213(88)90442-7.
- [2] F. Maltoni, K. Paul, T. Stelzer, and S. Willenbrock, *Color-flow decomposition of QCD amplitudes*, Phys. Rev. D **67** (2003) 014026, doi:10.1103/PhysRevD.67.014026, arXiv:hep-ph/0209271.
- [3] C. Degrande, C. Duhr, B. Fuks, D. Grellscheid, O. Mattelaer, and T. Reiter, *UFO—The Universal FeynRules Output*, Comput. Phys. Commun. **183** (2012) 1201–1214, doi:10.1016/j.cpc.2012.01.022, arXiv:1108.2040.
- [4] L. Darmé et al., *UFO 2.0—The Universal Feynman Output format*, Eur. Phys. J. C **83** (2023) 631, doi:10.1140/epjc/s10052-023-11780-9, arXiv:2304.09883.
- [5] R. Frederix and T. Vitos, *Event generation with exponential scaling in multiplicity using AmpliCol*, JHEP **04** (2026) 192, doi:10.1007/JHEP04(2026)192, arXiv:2601.19483.